CLASSIFICATION OF BIPARTITE OBSERVABLES III: SIMPLICITY, MIXTURES AND QUANTUM ERROR CORRECTION

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Abstract. We characterize logical Pauli errors in stabilizer codes through forking, the dependence of a type on additional parameters. Among errors undetectable by the code's checks, stabilizers act trivially on the encoded information and logical errors change it. An undetectable error on the code's qubits is logical exactly when its type over the stabilizer labels and a complete set of logical-operator labels forks over the stabilizer labels. The finite-support Pauli labels form a symplectic space whose theory, like that of the Rado graph, is simple and unstable; the independence theorem gives a compatibility rule for syndromes. Passing from labels to mixed preparations and probes changes the theory: for a diagonal observable of a two-valued relation on countable label sets, the independence property forces its mixed response theory to have the tree property of the second kind, TP_2, witnessed by an array with uniformly inconsistent rows and consistent choices of one condition per row. For the Pauli observable, actual mixtures parameterize a witnessing array for the response predicate itself. For the Rado observable, every TP_2 witnessing array for a fixed quantifier-free formula in a probe variable with only preparation parameters requires preparation parameters from an elementary extension outside the original mixed response structure.

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1. Introduction

We study how model-theoretic independence describes the algebra of stabilizer codes, and how allowing probabilistic mixtures changes the classification of the resulting response structures. Mixtures give uniform approximation of stable idealized responses in Part I and constrain the topology of full response compacta through convexity in Part II. Here we compare the simple theories of Pauli labels and graph vertices with the theories of their mixed response structures.

A stabilizer code protects quantum information by storing it in the common eigenvalue-1 space of commuting checks, which are Pauli strings: tensor products of identities, bit flips, phase flips and their products on individual qubits, the two-level systems of quantum information. A Pauli error may change the signs returned by the checks; the resulting list of signs, recorded as bits, is its syndrome. Errors with zero syndrome include both stabilizers, which act trivially on the code, and logical errors, which change the encoded information. We express this distinction through forking, a dependence relation for types, the collections of formulas satisfied by an element over a specified set of parameters. For an undetectable Pauli error, its type forks over the stabilizer labels after logical operators are added to the parameters exactly when the error is logical. The accompanying independence theorem gives a rule for combining compatible syndrome prescriptions. These statements concern dependence and compatibility in the label structure; the mixed preparations of the associated observable form a different structure, whose theory is no longer simple. Correctability is preserved under linear combinations of a specified set of error operators, so correcting a set of Pauli errors also corrects their linear span (see [Got97]).

To make this precise, let $\mathbb{F}_2$ be the two-element field and let n be a positive integer. The Hilbert space of a qubit is $\mathbb{C}^2$, on which the bit flip X and the phase flip Z, two of the Pauli matrices, exchange the two vectors of the standard basis and change the sign of the second, respectively. A Pauli string on n qubits is a tensor product of n factors $X^{x_k}Z^{z_k}$, for $k = 1, \dots, n$, with $x_k, z_k \in \mathbb{F}_2$, multiplied by one of the phases ± 1 and $\pm \mathrm{i}$; up to its phase, it is determined by its label, the vector $(x, z) \in \mathbb{F}_2^{2n}$ with $x = (x_1, \dots, x_n)$ and $z = (z_1, \dots, z_n)$. Let ω be the form on $\mathbb{F}_2^{2n}$ given by $\omega((x, z), (x', z')) = x \cdot z' + z \cdot x'$ for $x, z, x', z' \in \mathbb{F}_2^n$, where $x \cdot z' = \sum_{k=1}^{n} x_k z'_k$; it is bilinear, alternating and nondegenerate, and it is the standard symplectic form on $\mathbb{F}_2^{2n}$. Two Pauli strings on n qubits either commute or anticommute, that is, their products in the two orders are equal or opposite, and their labels decide which: the strings anticommute exactly when ω takes the value 1 at their labels. We disregard phases and identify each Pauli string with its label. Measuring a check g on a state that g fixes, after an error e, itself a Pauli string on the n qubits, has acted on the state, reveals whether the two strings anticommute: the syndrome bit of e with respect to g is 1 if they anticommute and 0 if they commute, and in terms of labels it is $\omega(g, e)$. The syndrome of e with respect to checks $g_1, \dots, g_m$, for a positive integer m, is the vector of its syndrome bits with respect to $g_1, \dots, g_m$.

On countably many qubits, the labels of Pauli strings with finite support form a vector space V of countably infinite dimension over $\mathbb{F}_2$; the same formula defines a nondegenerate alternating form ω on V. This symplectic label structure is the Pauli group modulo phases, with its vector-space operations and commutation form retained. In Section 3 we identify it as the Fraïssé limit of the finite alternating spaces over $\mathbb{F}_2$, meaning the countable homogeneous structure whose finite-dimensional substructures are exactly those spaces, including the degenerate ones. Its comparison with the Rado graph organizes the paper. The Rado graph is the countable homogeneous graph whose finite induced subgraphs include every finite graph;

it is also the almost-sure isomorphism type obtained by choosing each edge independently with probability one half [Fra54, Rad64, ER63]. Both theories are simple and unstable, as we prove in Theorems 3.1 and 3.3. Here simplicity excludes a formula and an infinitely branching tree of its instances with consistent branches and a fixed finite bound on how many children of any node can hold together; instability allows a formula to encode an infinite order pattern. For both structures we compute an independence relation over a specified base and verify the Kim–Pillay axioms, which characterize simplicity (see Fact 2.2). For the symplectic space, independence says that the spans generated by the two sets together with the base intersect only in the span of that base. This computation is the link from the infinite label structure to finite codes.

The dictionary in Section 4 makes this link explicit. For a stabilizer code on finitely many qubits, let S be the set of labels of its stabilizers, and choose logical-operator labels L so that $S \cup L$ spans the labels of all undetectable errors on those qubits. For every such error e, we prove that e is logical exactly when $\mathrm{tp}(e/S \cup L)$ forks over S (see Proposition 4.7). The base matters: all logical errors have the same type over S, and that type does not fork over the empty set. Forking here records the linear dependence introduced by the additional logical operators, rather than a distinction already visible in the syndrome alone. Clifford unitaries, which take Pauli strings to Pauli strings under conjugation up to phases, realize the homogeneity of the label structure: every isomorphism between its finite-dimensional substructures extends to an automorphism induced by a Clifford unitary on finitely many qubits, and every stabilizer code admits an encoding circuit of Clifford gates (see Theorem 4.3 and Corollary 4.4). The syndrome-gluing result concerns two sets of checks whose spans, after adjoining common checks, meet only in the span of those common checks. If two syndrome prescriptions are separately realized and agree on that intersection, a single error realizes both (see Proposition 4.8). This is a compatibility statement for prescribed linear functionals on check spans; it does not select a correcting error from measured syndrome data. We then compare the diagonal observables defined by the Rado relation and by the symplectic form, for which mixed preparations reveal a distinction absent from the classification of the label structures.

The passage to observables replaces a relation between labels by a real-valued response to probabilistic preparations and probes. A bipartite observable is a bounded self-adjoint operator on the tensor product of two Hilbert spaces. A density operator, a positive operator of trace one, describes Alice's actual preparation; averaging the observable against it gives an effective operator on Bob's space. These operators and the identity generate the probe algebra, the smallest norm-closed algebra containing them and their adjoints. A probe is a state of that algebra, a positive linear functional with value one at the identity, and the response is its value at the effective operator. In Part I [DI26a], we studied stability of this response function through agreement of the two orders of idealization of the preparations. For a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$, the ultralimit of a bounded sequence is the unique value approximated within every prescribed error on a set of indices belonging to $\mathcal{U}$; applying this to expectations gives a state of the bounded-operator algebra, which need not be represented by a density operator. In Part II [DI26b], we studied NIP for Alice's responses: the absence of an infinite sequence of her preparations on which probes realize every finite high/low pattern with a fixed positive gap. The direction is specified because Alice's preparations are the functions and Bob's probes their arguments. With Bob idealized first, stability means continuity of every idealized response on the probe space, whereas NIP means that every such response is of the first Baire class, a pointwise limit of a sequence of continuous functions. For a NIP observable these functions form a separable Rosenthal compactum,

to which Todorčević's trichotomy applies. Simplicity concerns a different question: whether instances of a single formula can form an infinitely branching tree with consistent branches and some fixed integer $k \geq 2$ such that no k distinct children of any node are jointly consistent. As properties of complete first-order theories, simplicity and NIP do not imply one another, and their intersection is stability. For the code interpretation the unknown element is an error and the conditions concern its syndrome; for the observable structure the unknown element is a probe and the conditions are inequalities between responses.

In Section 5 we associate diagonal observables with the two label structures. Alice's and Bob's Hilbert spaces have orthonormal bases indexed by countable sets of labels, and the observable records whether the relation holds between their labels. For the Rado observable it records adjacency of two vertices; for the Pauli observable Alice's label is a check, Bob's is an error, and it records anticommutation. Only the diagonal of Alice's density operator enters the response, so a mixed preparation is represented by a probability measure on her labels, and the response is affine in that measure. We call the point masses pure preparations in this label description. The probe spaces retain a distinction between the two relations. For the Rado observable they are Radon probability measures on the Cantor cube of all vertex subsets, with pure probes given by individual subsets, so every pattern of answers to Alice's labels is permitted (see Lemma 5.5). For the Pauli observable they are probability measures on the compact product group of Pauli labels of arbitrary support, and its pure probes are individual labels in that group (see Proposition 5.4). Each such label prescribes syndrome bits on all finite-support checks, with the prescribed bits constrained by linearity. These probability measures are the Pauli channels of the present model; the syndrome sector of finitely many checks consists of the arbitrary-support labels with a prescribed syndrome. Independent depolarizing noise of rate p, for $0 < p \leq 1$, assigns the identity label probability $1 - p$ and each of the three nonidentity one-qubit labels probability $p/3$ at every qubit. Its product law is concentrated on infinite-support errors, outside V. Such an error is idealized relative to the finite-support label structure, but its point mass already belongs to the probe space of the observable; it need not be a new probe in an elementary extension.

The real-valued theory of an observable is the theory of the structure whose two sorts are Alice's mixed preparations and Bob's probes, with the response as predicate, in Keisler's real-valued logic [Kei23]. We prove that the theory of every diagonal observable with the independence property has the tree property of the second kind, abbreviated TP_2 (see Theorem 6.2). This property is witnessed by an array of conditions on a probe: each row is uniformly k-inconsistent for some fixed integer $k \geq 2$, while every choice of one condition from each row is consistent. Uniform inconsistency is expressed by a fixed positive gap, as specified in Definition 5.2. The witnessing formula is the difference of two responses. On a finite set of labels, its value for the uniform mixtures on a half and its complement reaches 1 exactly when the probe answers 1 on the half and 0 on the complement; two halves meeting in a quarter of the set give differences that cannot both exceed $1/2$. The independence property supplies the compatible finite paths, and compactness gives the array in an elementary extension. Thus the simple unstable theories of the original label structures do not describe their mixed response structures. The location of the array parameters then separates the Pauli and Rado observables. For the Pauli observable, actual mixtures parameterize a TP_2 array for the response predicate itself; every finite part of a path has a finite-support error realizing it, while a whole path is realized by an arbitrary-support pure probe already in the original probe space (see Theorem 7.2). For the Rado observable, no fixed quantifier-free formula on probes with only actual preparations as parameters admits such an array, so any witnessing array of this kind must contain preparation parameters from an elementary extension (see

Theorem 8.3 and Corollary 8.2). In this discussion of preparation parameters, we call a parameter "actual" when it belongs to the original mixed-preparation sort and "idealized" when it is a new preparation parameter in an elementary extension of the mixed response structure. In an ultrapower along a nonprincipal ultrafilter, the response of two represented sequences is the ultralimit of their responses, and Łoś's theorem preserves the values of all formulas at the original parameters. The comparison with direct-integral mixtures of structures in affine logic is made in Section 9, followed by the questions in Section 10.

Theorem 1.1 (Main results).

(A) *The Rado graph and the symplectic space of countably infinite dimension over $\mathbb{F}_2$, which is the Pauli group on countably many qubits modulo phases, have simple unstable theories (Theorems 3.1 and 3.3).*

(B) *Let n be a positive integer, let a stabilizer code on n qubits be given, let S be the set of labels of its stabilizer group, and let L be a set of logical operators of the code, labels that together with S span the labels of its undetectable errors; labels on n qubits are regarded as elements of the Pauli group through the first n of countably many qubits. An undetectable error is a logical error exactly when its type over $S \cup L$, in the theory of the Pauli group, forks over S; all logical errors have the same type over S, and this type does not fork over the empty set (Theorem 4.5 and Proposition 4.7). Every isomorphism between finite-dimensional substructures of the Pauli group extends to an automorphism induced by a Clifford unitary on finitely many qubits, and every stabilizer code has an encoding circuit made of Clifford gates (Theorem 4.3 and Corollary 4.4). The independence theorem becomes the gluing of syndromes: if two sets of checks are independent over a common set, then any two syndromes, each realized by some error on its own set of checks, that agree on the common set are realized by a single error (Proposition 4.8). The probes of the Pauli observable are the Pauli channels, and the depolarizing noise of every positive rate on countably many qubits is concentrated on the Pauli strings of infinite support, which are idealized errors relative to the finite-support label structure (Proposition 5.4).*

(C) *For every diagonal observable whose pure responses, the responses of single labels, have the independence property, and in particular for the Rado observable and the Pauli observable, the real-valued theory of the observable has the tree property of the second kind, for the formula that is the difference of two responses. Since the independence property of a diagonal observable passes to its pure responses, it follows that a diagonal observable whose real-valued theory is simple is NIP (Theorem 6.2).*

(D) *For the Pauli observable, the indicator of the syndrome sector of every nonzero syndrome of finitely many linearly independent checks is, on pure probes, the difference of the responses of two uniform mixtures of checks, and the response itself has the tree property of the second kind, witnessed by actual preparations: for blocks of checks that are linearly independent all together, the conditions that a given check is the first of its block to fire, that is, to anticommute with the unknown error, form the rows of a witnessing array; in the bit-flip repetition code, whose checks compare neighboring qubits of a chain, the first check to fire marks the first domain wall of a bit flip, the first boundary between flipped and unflipped qubits (Proposition 7.1, Theorem 7.2 and Example 7.4).*

(E) *For the Rado observable, and more generally for every diagonal observable with freely independent pure responses, that is, one in which every pattern of answers to Alice's*

labels is a pure probe, such as the observable of the random bipartite graph, the Fraïssé limit of the finite bipartite graphs with the two sides as two sorts, the conditions on a probe given by instances of a quantifier-free formula with actual mixtures as parameters are compatible along every sequence: no such sequence is uniformly k-inconsistent, for any integer $k \geq 2$. Hence, for these observables, every array that witnesses the tree property of the second kind of the formula in (C) contains idealized preparations (Theorems 8.1 and 8.3, and Corollary 8.2).

The distinction between (D) and (E) comes from the relations among pure responses. For the Pauli observable, the arbitrary-support error labels form a compact group, and Alice's pure responses become its characters after an affine change of scale, where a character is a continuous homomorphism into $\{\pm 1\}$ (see Proposition 5.4). Products of characters remain characters, so averaging over the finite group generated by independent checks expresses the indicator of a nonzero syndrome sector, on pure probes, as a difference of two uniform-mixture responses (see Proposition 7.1). This identity parallels the stabilizer expression for a syndrome projector [Got97], but requires linear independence of the checks rather than their commutation. A uniform mixture on the coset of one check modulo the span of its predecessors consequently has response 1 at a pure probe exactly when that check is the first in its block to anticommute with the error. This supplies the actual preparation parameters in (D). For the Rado observable, pure responses are coordinate functions on the whole Cantor cube, and responses to actual mixtures depend on a probe only through its coordinate marginals. In Theorem 8.1, a gliding-hump argument separates the weights of a sequence of bounded affine combinations of responses into a common finite part and nearly disjoint finite blocks, allowing one probe to nearly maximize any prescribed finite number of a suitable subsequence. The bound is on the sum of the absolute values of the constant term and coefficients, as in that theorem. For a fixed quantifier-free formula, a continuous function of finitely many responses, the corresponding argument glues maximizers on disjoint blocks and excludes uniformly k-inconsistent sequences with actual preparation parameters for every integer $k \geq 2$ (see Theorem 8.3). The Pauli case therefore detects linear compatibility among syndrome bits through actual mixtures, while, for the stated quantifier-free formulas in a probe variable, the Rado case forces preparation parameters of a witnessing array beyond the original mixed-preparation sort.

2. Simple theories and the Kim–Pillay theorem

The independence theorem addresses a compatibility problem: two nonforking extensions of the same type over a model, to parameter sets independent over that model, admit a common nonforking extension. For Pauli labels, types specify syndrome bits and linear relations, and this compatibility is expressed by the symplectic form and linear span. We first recall the model-theoretic formulation, including the tree configurations excluded by simplicity and the Kim–Pillay characterization by an independence relation. In Section 3 we verify this characterization for the Rado graph and the symplectic space of Pauli labels.

To fix the setting, we work in a monster model $\mathbb{M}$ of a complete first-order theory T: a model of very large cardinality in which every set of formulas in finitely many variables with parameters from a subset of smaller cardinality is satisfied whenever it is finitely satisfiable, and in which every elementary map between two such subsets, that is, every map under which each tuple and its image satisfy the same formulas, extends to an automorphism. These subsets are called small, and from now on all sets of parameters are assumed to be

small. The type $\mathrm{tp}(a/C)$ of a tuple a over a set C is the set of formulas with parameters from C that a satisfies. For sets B and D we write BD for $B \cup D$, and, identifying a tuple a with the set of its entries, we write aB for the union of that set with B. For a finite sequence $s \in \mathbb{N}^{<\mathbb{N}}$ of natural numbers and $j \in \mathbb{N}$, we write $s^\frown j$ for the sequence obtained by appending j to s, and for an infinite sequence $\eta \in \mathbb{N}^{\mathbb{N}}$ and $n \in \mathbb{N}$, we write $\eta \restriction n$ for the sequence of the first n terms of η. Two finite sequences are incomparable when neither is an initial segment of the other.

Definition 2.1. Let $\varphi(x, y)$ be a formula, where x and y are finite tuples of variables.

(1) For an integer $k \geq 2$, a sequence $(b_j)_{j\in\mathbb{N}}$ of tuples of $\mathbb{M}$ of the length of y is *k-inconsistent for φ* if for all $j_1 < \cdots < j_k$ in $\mathbb{N}$ there is no x with $\varphi(x, b_{j_1}) \wedge \cdots \wedge \varphi(x, b_{j_k})$.
(2) The formula φ has the *tree property* if, for some $k \geq 2$, there is a tree $(b_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ such that for every $\eta \in \mathbb{N}^{\mathbb{N}}$ the set $\{\varphi(x, b_{\eta\restriction n}) : n \in \mathbb{N}\}$ is consistent and for every s the sequence $(b_{s^\frown j})_{j\in\mathbb{N}}$ is k-inconsistent for φ.
(3) The formula φ has the *tree property of the first kind* if there is a tree $(b_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ such that for every $\eta \in \mathbb{N}^{\mathbb{N}}$ the set $\{\varphi(x, b_{\eta\restriction n}) : n \in \mathbb{N}\}$ is consistent and for all incomparable $s, t \in \mathbb{N}^{<\mathbb{N}}$ the set $\{\varphi(x, b_s), \varphi(x, b_t)\}$ is inconsistent.
(4) The formula φ has the *tree property of the second kind* (TP_2) if, for some $k \geq 2$, there is an array $(b_{ij})_{i,j\in\mathbb{N}}$ such that every row $(b_{ij})_{j\in\mathbb{N}}$ is k-inconsistent for φ and for every $f\colon \mathbb{N} \to \mathbb{N}$ the set $\{\varphi(x, b_{if(i)}) : i \in \mathbb{N}\}$ is consistent.
(5) The theory T is *simple* if no formula has the tree property.

Recall that the theory T is stable when no formula $\varphi(x, y)$ has the order property, that is, when there are no tuples a_i and b_i, for $i \in \mathbb{N}$, such that, for all $i, j \in \mathbb{N}$, the formula $\varphi(a_i, b_j)$ holds exactly when $i < j$. It is NIP when no formula $\varphi(x, y)$ has the independence property, that is, when there are no tuples b_i, for $i \in \mathbb{N}$, such that every subset of $\mathbb{N}$ has the form $\{i \in \mathbb{N} : \varphi(a, b_i) \text{ holds}\}$ for some tuple a. Every stable theory is simple, and the tree property of the second kind implies the tree property; by a theorem of Shelah, a theory is simple exactly when it has neither TP_2 nor the tree property of the first kind (see [She90, Theorem III.7.11]). A simple theory is stable if and only if it is NIP (see [She80, GIL02]), so a simple unstable theory has the independence property. To prove simplicity we do not build trees; instead, we exhibit an independence relation directly and show that it is nonforking independence, which we now recall. For a formula $\varphi(x, y)$ and a tuple b of $\mathbb{M}$ of the length of y, the formula $\varphi(x, b)$ *divides* over a set C when, for some integer $k \geq 2$, there is a sequence $(b_j)_{j\in\mathbb{N}}$ with $b_0 = b$ that is k-inconsistent for φ and indiscernible over C, that is, such that any two of its increasing finite subsequences of the same length have the same type over C. The formula *forks* over C when it implies a finite disjunction of formulas each of which divides over C; a type forks over C when one of its formulas does. Nonforking independence is the relation under which $A \mathop{\smile\!\!\!\!\!\!|}_C B$ holds, for small sets A, B and C, when, for every finite tuple a from A, the type $\mathrm{tp}(a/BC)$ does not fork over C. By the Kim–Pillay theorem (see Fact 2.2), every relation with the properties listed there coincides with nonforking independence.

Fact 2.2 (Kim and Pillay). *Let $\mathop{\smile\!\!\!\!\!\!|}$ be a ternary relation on small subsets of $\mathbb{M}$, written $A \mathop{\smile\!\!\!\!\!\!|}_C B$, and assume that it satisfies the following:*

(i) invariance *under automorphisms of $\mathbb{M}$;*
(ii) symmetry*: the relation $A \mathop{\smile\!\!\!\!\!\!|}_C B$ holds if and only if $B \mathop{\smile\!\!\!\!\!\!|}_C A$ holds;*
(iii) monotonicity and transitivity*: the relation $A \mathop{\smile\!\!\!\!\!\!|}_C BD$ holds if and only if both $A \mathop{\smile\!\!\!\!\!\!|}_C B$ and $A \mathop{\smile\!\!\!\!\!\!|}_{CB} D$ hold;*
(iv) finite character*: the relation $A \mathop{\smile\!\!\!\!\!\!|}_C B$ holds if and only if $a \mathop{\smile\!\!\!\!\!\!|}_C b$ holds for all finite tuples a from A and b from B;*

(v) local character*: for every finite tuple a and every set B there is $B_0 \subseteq B$ with $|B_0| \leq |T|$ and $a \mathop{\smile\!\!\!\!\!\!|}_{B_0} B$, where $|T|$ is the number of formulas of the language of T;*
(vi) extension*: for every tuple a and all sets $C \subseteq B$ there is a' with $\mathrm{tp}(a'/C) = \mathrm{tp}(a/C)$ and $a' \mathop{\smile\!\!\!\!\!\!|}_{C} B$;*
(vii) independence theorem over models*: if M is a small elementary substructure of $\mathbb{M}$, $B \mathop{\smile\!\!\!\!\!\!|}_{M} D$, $a_1 \mathop{\smile\!\!\!\!\!\!|}_{M} B$, $a_2 \mathop{\smile\!\!\!\!\!\!|}_{M} D$ and $\mathrm{tp}(a_1/M) = \mathrm{tp}(a_2/M)$, then there is a with $\mathrm{tp}(a/MB) = \mathrm{tp}(a_1/MB)$, $\mathrm{tp}(a/MD) = \mathrm{tp}(a_2/MD)$ and $a \mathop{\smile\!\!\!\!\!\!|}_{M} BD$.*

Then the theory T is simple and the relation $\mathop{\smile\!\!\!\!\!\!|}$ coincides with nonforking independence (see [KP97, Theorem 4.2]*).*

3. The two paradigms

The Rado graph and the Pauli labels supply two concrete realizations of the independence calculus. In each case we identify a Fraïssé limit and use quantifier elimination to describe types by graph relations and linear-algebraic data, respectively. We prove that both theories are simple and unstable, with algebraic closure equal to the set itself in the graph and to linear span in the Pauli structure. We shall then compare the theories of their mixed response structures, whose different behavior depends on the relations among the labels.

3.1. **The Rado graph.** The class of finite graphs has the amalgamation property, since two graphs that share an induced subgraph can be glued along it without new edges. Its Fraïssé limit is the Rado graph (Γ, E), with vertex set Γ and edge relation E, and a countable graph is isomorphic to the Rado graph if and only if it has the *extension property*: for all finite disjoint sets Γ_1 and Γ_0 of vertices, there are infinitely many vertices outside $\Gamma_1 \cup \Gamma_0$ adjacent to every vertex of Γ_1 and to no vertex of Γ_0 [Fra54, Rad64]. The theory of (Γ, E) is $\aleph_0$-categorical and eliminates quantifiers, so the type of a tuple over a set D is determined by the equalities and edges among its entries and between its entries and the elements of D. By saturation, the monster model $\mathbb{M}$ of the theory of (Γ, E) has the extension property for small disjoint sets Γ_1 and Γ_0 of vertices.

The relation $E(x, y)$ has the independence property, since, by compactness, it suffices to realize every finite pattern: for every positive integer n, all distinct vertices $a_1, \ldots, a_n$ of the Rado graph and every set of indices among $1, \ldots, n$, the extension property gives a vertex adjacent to a_i exactly when i is in the set. Hence the theory of the Rado graph is unstable.

Theorem 3.1. *The theory of the Rado graph is simple, and, for small sets A, B and C of vertices of the monster model $\mathbb{M}$ of this theory, nonforking independence $A \mathop{\smile\!\!\!\!\!\!|}_{C} B$ holds if and only if $A \cap B \subseteq C$.*

Proof. We verify the hypotheses of Fact 2.2 for the relation $A \mathop{\smile\!\!\!\!\!\!|}_{C} B$ defined by $A \cap B \subseteq C$. The first five are set algebra, and extension and the independence theorem follow from the extension property of $\mathbb{M}$. Invariance, symmetry and finite character are immediate. Monotonicity and transitivity hold together, since the conditions $A \cap B \subseteq C$ and $A \cap D \subseteq C \cup B$ say jointly that every element of $A \cap (B \cup D)$ lies in C. For local character, given a finite tuple a and a set B, the finite set $B_0 = B \cap a$ satisfies $a \cap B \subseteq B_0$, which is the relation $a \mathop{\smile\!\!\!\!\!\!|}_{B_0} B$.

In the two constructions below, we temporarily group the coordinates whose entries lie in the base before the remaining coordinates, and restore the original order at the end. For two tuples with the same type over the base, use the same coordinate permutation. Repeated entries retain their equality pattern: one new vertex is chosen for each distinct entry outside the base. For extension, let a and $C \subseteq B$ be given, and write $a = (c, a_0)$, where c lists the

entries of a in C. The type of a_0 over C prescribes only the equalities and edges of a_0 within itself and to C. Hence the extension property of $\mathbb{M}$, applied to the distinct entries in turn, with $B \setminus C$ added to the set of vertices to which nonadjacency is prescribed, gives a tuple a_0' disjoint from B with the same type over C, and the tuple $a' = (c, a_0')$ satisfies $a' \cap B \subseteq C$.

For the independence theorem, let M, B, D, a_1 and a_2 be as in Fact 2.2(vii). Since $\mathrm{tp}(a_1/M) = \mathrm{tp}(a_2/M)$, the two tuples have the same entries in M and the same remaining equalities; write $a_\ell = (m, a_\ell')$ for $\ell = 1, 2$, where the tuple m lists these common entries and each a_ℓ' is disjoint from M. The hypotheses $a_1 \mathop{\smile\!\!\!\!\!\!|}_M B$ and $a_2 \mathop{\smile\!\!\!\!\!\!|}_M D$ say that a_1' is disjoint from B and a_2' from D. We look for a tuple x, disjoint from $M \cup B \cup D$, whose equalities and edges within itself and edges to M are those of a_1' (equivalently, of a_2'), whose edges to $B \setminus M$ are those of a_1', and whose edges to $D \setminus M$ are those of a_2'. These prescriptions concern disjoint sets of vertices, because $B \cap D \subseteq M$; the extension property of $\mathbb{M}$ therefore provides such an x, and by quantifier elimination the tuple $a = (m, x)$ is as required. □

3.2. **Pauli strings.** The Hilbert space of a qubit is $\mathbb{C}^2$, and the bit flip and the phase flip are two of the Pauli matrices, which anticommute:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad XZ = -ZX.$$

For a positive integer n, a Pauli string on n qubits is, up to a phase, a tensor product $X^{x_1}Z^{z_1} \otimes \cdots \otimes X^{x_n}Z^{z_n}$ with x_k and z_k in the two-element field $\mathbb{F}_2$ for $k = 1, \ldots, n$, and it is labeled by the vector $(x, z) \in \mathbb{F}_2^{2n}$, where $x = (x_1, \ldots, x_n)$ and $z = (z_1, \ldots, z_n)$; products of strings correspond to sums of labels. For a label $v = (x, z)$ we write P_v for the Hermitian string $\mathrm{i}^{N(v)}X^{x_1}Z^{z_1} \otimes \cdots \otimes X^{x_n}Z^{z_n}$, where $N(v)$ is the number of indices k with $x_k = z_k = 1$; it satisfies $P_v^2 = I$, where I is the identity, and the label $(1, 1)$ of one qubit gives the third Pauli matrix $P_{(1,1)} = \mathrm{i}XZ = Y$. Two strings either commute or anticommute, and

$$P_{(x,z)}P_{(x',z')} = (-1)^{\omega((x,z),(x',z'))}P_{(x',z')}P_{(x,z)}, \qquad \omega((x, z), (x', z')) = x \cdot z' + z \cdot x',$$

where $x \cdot z' = \sum_{k=1}^{n} x_k z_k'$, computed in $\mathbb{F}_2$, as in the Introduction. The form ω is bilinear, alternating and nondegenerate on $\mathbb{F}_2^{2n}$. Recall that in error correction an error is a Pauli string that acts on an encoded state as noise, and a check is a Pauli string whose measurement detects errors; as in the Introduction, we identify each Pauli string, up to its phase, with its label. The *syndrome* of an error e with respect to checks $g_1, \ldots, g_m$, for a positive integer m, is the vector $(\omega(g_1, e), \ldots, \omega(g_m, e))$: when the checks commute pairwise, measuring them on a state that they fix, after the error has acted on it, reveals which of them anticommute with the error. On countably many qubits we write V for the direct sum $\bigoplus_{k \in \mathbb{N}} \mathbb{F}_2^2$, with the form given by the same formula, now a finite sum for each pair of vectors; its elements are the labels of the Pauli strings that act nontrivially on only finitely many qubits.

3.3. **The Pauli group as a Fraïssé limit.** In the language with $+$, 0 and a binary relation for $\omega(x, y) = 1$, substructures are $\mathbb{F}_2$-subspaces with the restricted form. Finite alternating spaces over $\mathbb{F}_2$, the finite vector spaces over $\mathbb{F}_2$ with an alternating bilinear form, amalgamate: if $W_0 \subseteq W_1$ and $W_0 \subseteq W_2$, write $W_1 = W_0 \oplus W_1'$ and $W_2 = W_0 \oplus W_2'$, and give $W_0 \oplus W_1' \oplus W_2'$ the bilinear form that extends the forms of W_1 and W_2 and vanishes on $W_1' \times W_2'$ and on $W_2' \times W_1'$. Every finite alternating space embeds in a nondegenerate one, and by Witt's theorem [Art57] every isometry between subspaces of a finite nondegenerate alternating space extends to an isometry of the whole space. A back-and-forth argument along symplectic bases therefore shows that (V, ω) is the Fraïssé limit of the finite alternating spaces over $\mathbb{F}_2$. Its theory is $\aleph_0$-categorical and eliminates quantifiers, so the type of a tuple a over a set D is determined

by the linear relations of a over $\mathrm{span}(D)$ and the values of ω on $\mathrm{span}(aD)$. The algebraic closure of a set is its span.

Lemma 3.2. *Let n be a positive integer. If $a_1, \ldots, a_n \in V$ are linearly independent and $\varepsilon_1, \ldots, \varepsilon_n \in \mathbb{F}_2$, then there is $b \in V$ with $\omega(a_i, b) = \varepsilon_i$ for $i = 1, \ldots, n$.*

Proof. The vectors a_i lie in the span W of the labels of X and Z on the first m qubits, for a large enough integer m, and the form is nondegenerate on W. Hence the functionals $\omega(a_i, \cdot)$ restricted to W are linearly independent, and the linear map $b \mapsto (\omega(a_i, b))_{i \leq n}$ from W to $\mathbb{F}_2^n$ is onto. □

By Lemma 3.2 and compactness, the formula $\omega(x, y) = 1$ has the independence property: every syndrome with respect to finitely many linearly independent checks is the syndrome of some error. Hence the theory of (V, ω) is unstable. We shall also use the finite-extension property of the Fraïssé limit. Every finite alternating extension of a finite subspace of V embeds into V over that subspace: embed the extension into V, then use homogeneity to match the given copy of the subspace. For each pair of finite alternating spaces, this property is expressed by a first-order extension sentence, so it holds in the monster model $\mathbb{M}$ as well. Consequently, for every small subspace D of $\mathbb{M}$ and every positive integer n, the conditions describing an alternating extension $D \oplus \mathrm{span}(x_1, \ldots, x_n)$ by n independent vectors are finitely satisfiable: each finite set of conditions concerns only a finite-dimensional subspace of D and a finite alternating extension of it. Saturation realizes these conditions over D. In particular, every linear functional f on D has the form $\omega(\cdot, b)$ on D for some $b \in \mathbb{M}$.

Theorem 3.3. *The theory of (V, ω) is simple, and, for small subsets A, B and C of the monster model $\mathbb{M}$ of this theory, nonforking independence $A \mathop{\smile\!\!\!\!\!\!|}_C B$ holds if and only if $\mathrm{span}(AC) \cap \mathrm{span}(BC) = \mathrm{span}(C)$.*

Proof. The proof follows that of Theorem 3.1, with linear spans in place of the sets themselves: the first five axioms are linear algebra, and extension and the independence theorem come from the freedom to prescribe the form on a complement. We verify the hypotheses of Fact 2.2 for the relation $A \mathop{\smile\!\!\!\!\!\!|}_C B$ defined by $\mathrm{span}(AC) \cap \mathrm{span}(BC) = \mathrm{span}(C)$. Invariance and symmetry are immediate, and finite character holds because a vector of $\mathrm{span}(AC) \cap \mathrm{span}(BC)$ outside $\mathrm{span}(C)$ is a combination of finitely many elements of A, B and C.

For monotonicity and transitivity, put $Q_1 = \mathrm{span}(AC)$, $Q_2 = \mathrm{span}(BC)$, $Q_3 = \mathrm{span}(BCD)$ and $Q_0 = \mathrm{span}(C)$; these subspaces satisfy $Q_2 \subseteq Q_3$ and $\mathrm{span}(ABC) = Q_1 + Q_2$. It remains to be shown that $Q_1 \cap Q_3 = Q_0$ if and only if $Q_1 \cap Q_2 = Q_0$ and $(Q_1 + Q_2) \cap Q_3 = Q_2$. Suppose that $Q_1 \cap Q_3 = Q_0$. Since $Q_2 \subseteq Q_3$, the intersection $Q_1 \cap Q_2$ is contained in Q_0. If $u_1 + u_2 \in Q_3$ with $u_1 \in Q_1$ and $u_2 \in Q_2$, then the vector u_1 also lies in Q_3, hence in $Q_1 \cap Q_3 = Q_0 \subseteq Q_2$; thus the sum lies in Q_2. Conversely, every $u \in Q_1 \cap Q_3$ lies in $(Q_1 + Q_2) \cap Q_3 = Q_2$, and therefore in $Q_1 \cap Q_2 = Q_0$.

Local character follows from finite dimension. Given a finite tuple a and a set B, the space $\mathrm{span}(a) \cap \mathrm{span}(B)$ is finite, so it is contained in $\mathrm{span}(B_0)$ for some finite $B_0 \subseteq B$. If $u + w \in \mathrm{span}(B)$ with $u \in \mathrm{span}(a)$ and $w \in \mathrm{span}(B_0)$, then the vector u lies in $\mathrm{span}(a) \cap \mathrm{span}(B) \subseteq \mathrm{span}(B_0)$. It follows that $\mathrm{span}(aB_0) \cap \mathrm{span}(B) = \mathrm{span}(B_0)$, which is the relation $a \mathop{\smile\!\!\!\!\!\!|}_{B_0} B$.

For extension and the independence theorem, we first treat finite tuples. For a small tuple, each finite fragment of the required type and of the span-independence conditions involves only finitely many coordinates; the finite-tuple constructions therefore make these conditions finitely satisfiable, and compactness and saturation realize them together. In either construction, choose a maximal subtuple linearly independent over the base span. Every

omitted coordinate is a linear combination of the chosen coordinates and base parameters, and is restored by the same equation after the construction. In the independence theorem, equality of the two types over the base ensures that the same coordinate indices can be chosen and that these restoring equations agree. For extension, let a and $C \subseteq B$ be given, with a now linearly independent over $\mathrm{span}(C)$. Choose a complement B' of $\mathrm{span}(C)$ in $\mathrm{span}(BC)$. By the realization property established above, for a tuple x of new vectors of the same length as a, the alternating space $\mathrm{span}(BC) \oplus \mathrm{span}(x)$ in which x has the values of ω prescribed by $\mathrm{tp}(a/C)$ on $\mathrm{span}(xC)$ and is orthogonal to B' is realized over $\mathrm{span}(BC)$ by a tuple a'; this tuple has the type of a over C and is linearly independent over $\mathrm{span}(BC)$, whence $\mathrm{span}(a'C) \cap \mathrm{span}(BC) = \mathrm{span}(C)$, which is the relation $a' \mathop{\smile\!\!\!\!\!\!|}_C B$.

The independence theorem reduces to a gluing of functionals. Let M, B, D, a_1 and a_2 be as in Fact 2.2(vii). As in the proof of extension, we may assume that a_1 is linearly independent over M; then so is a_2, and the hypotheses $a_1 \mathop{\smile\!\!\!\!\!\!|}_M B$ and $a_2 \mathop{\smile\!\!\!\!\!\!|}_M D$ make a_1 linearly independent over $\mathrm{span}(MB)$ and a_2 over $\mathrm{span}(MD)$. Let n be the common length of a_1 and a_2, and, for $\ell = 1, 2$ and $k = 1, \ldots, n$, let $a_{\ell,k}$ be the kth entry of a_ℓ. For each k, the functionals $\omega(a_{1,k}, \cdot)$ on $\mathrm{span}(MB)$ and $\omega(a_{2,k}, \cdot)$ on $\mathrm{span}(MD)$ agree on M, because $\mathrm{tp}(a_1/M) = \mathrm{tp}(a_2/M)$. Since $B \mathop{\smile\!\!\!\!\!\!|}_M D$, their domains meet in $\mathrm{span}(MB) \cap \mathrm{span}(MD) = M$, and the two functionals glue to a linear functional on $\mathrm{span}(MBD)$. Realize over $\mathrm{span}(MBD)$ a tuple a, linearly independent over $\mathrm{span}(MBD)$, with these functionals and with the values of ω on a given by $\mathrm{tp}(a_1/M)$. By quantifier elimination, the type of a over MB is that of a_1, and its type over MD is that of a_2; the relation $a \mathop{\smile\!\!\!\!\!\!|}_M BD$ holds by construction. □

In both structures, then, nonforking independence is the independence given by algebraic closure: for small sets A, B and C, the relation $A \mathop{\smile\!\!\!\!\!\!|}_C B$ holds exactly when the closures of AC and BC meet in the closure of C. The independence theorem then reduces to the freedom to prescribe the remaining relations, edges or values of the form, on each side separately.

Remark 3.4. The proof of Theorem 3.3 applies over every finite field $\mathbb{F}_q$, for a prime power q, in the language with a unary function for multiplication by each scalar and a binary relation for each value of the form. For symmetric forms in characteristic two, and for arbitrary bilinear forms, the identification of the limit through symplectic bases and Witt's theorem is no longer available, but the Fraïssé limit still exists, since the finite spaces with such a form have free amalgamation. The verification of Fact 2.2 goes through with the values $\omega(x, x)$ prescribed by the type and, for forms that are not symmetric, with the functionals $\omega(x, \cdot)$ and $\omega(\cdot, x)$ glued separately.

4. Stabilizer codes as a structure: a dictionary

The dictionary rests on the linear algebra of Pauli labels: multiplication modulo phases is vector addition, commutation is a symplectic pairing, and a syndrome is the restriction of that pairing to the span of the checks. These identifications connect three model-theoretic constructions with stabilizer codes. Finite-substructure homogeneity is expressed by Clifford conjugation, forking of a single label is detected by a change of linear span over a specified base, and the independence theorem has a finite-dimensional counterpart in the compatibility of syndrome functionals. After recalling stabilizer codes, we state and prove the corresponding results and give two readings, translating model theory for the physicist and stabilizer-code algebra for the logician. The Laflamme–Miquel–Paz–Zurek five-qubit code [LMPZ96], independently found by Bennett, DiVincenzo, Smolin and Wootters [BDSW96], is the running example; it is the smallest code correcting an arbitrary error on one qubit. The translations

are collected in Tables 1 and 2: first the label structure and its codes, then the response structure with mixtures and probes.

The first table concerns the symplectic label structure (V, ω) of Section 3, where $V = \bigoplus_{j\in\mathbb{N}} \mathbb{F}_2^2$ consists of finite-support labels and Pauli phases are discarded. For a positive integer n, fix a code on n qubits with stabilizer label space $S \subseteq \mathbb{F}_2^{2n}$, and put $S^\perp = \{v \in \mathbb{F}_2^{2n} : \omega(s, v) = 0 \text{ for all } s \in S\}$, with the perpendicular taken in this finite ambient space. Choose logical labels L with $S^\perp = S \oplus \operatorname{span}(L)$; in the two code-specific forking rows, $e \in S^\perp \setminus \{0\}$ is a nonzero error label with zero syndrome. In the general type rows, $D \subseteq B$ are parameter sets in a sufficiently saturated elementary extension of the label structure, and e is one label in that extension. The gluing row assumes that the two check spans intersect exactly in their common base span.

Model theory	**Label algebra and codes**	**Result**
(V, ω)	finite-support Pauli labels, addition and commutation	Section 3
finite substructures	alternating spaces; commutation and linear relations	Proposition 4.2
finite-substructure homogeneity	isometries extend by finite Clifford conjugation	Theorem 4.3
isotropic subspaces, with generator signs	stabilizer codes and Clifford encoding	Corollary 4.4
algebraic closure of B	$\operatorname{span}(B)$: generated strings modulo phases	Section 3
$\operatorname{tp}(e/B)$	syndrome values and product relations over B	Theorem 4.5
$\operatorname{tp}(e/B)$ forks over D	$e \in \operatorname{span}(B) \setminus \operatorname{span}(D)$	Theorem 4.5
$\operatorname{tp}(e/S)$ does not fork over $\emptyset$	$e \notin S$: a logical error	Corollary 4.6
$\operatorname{tp}(e/S \cup L)$ forks over S	$e \notin S$: the same logical error, with L added	Proposition 4.7
independence theorem: finite counterpart	syndrome functionals agreeing on the base glue	Proposition 4.8

TABLE 1. From Pauli labels to stabilizer codes. The two code-specific forking rows use the same nonzero error with zero syndrome and display the different parameter sets and bases.

The second table concerns the real-valued structure $\mathcal{S}_\varphi$ of preparations and probes from Section 5, specialized to the Pauli predicate $\varphi = \omega$. Thus $\mathcal{S}_\omega$ has probability distributions on V as preparations, and Radon probability measures on $\hat{V} = \prod_{j\in\mathbb{N}} \mathbb{F}_2^2$ as probes. The compact space $\hat{V}$ consists of arbitrary-support error labels and contains V; its point masses are the pure probes, with labels in V giving actual finite-support errors. The symplectic pairing of a finite-support check with any label in $\hat{V}$ is a finite sum, so every response is defined (see Proposition 5.4). A specified nonzero syndrome is a linear functional on the finite check span; its kernel has two cosets, labeled by the values 0 and 1.

4.1. **Stabilizer codes in brief.** We recall what the logician needs (see [Got97]). For a positive integer n, a *stabilizer group* on n qubits is a group $\mathcal{G}$ of pairwise commuting Pauli strings that does not contain $-I$. Its common eigenspace for the eigenvalue 1 is the *code space*;

Response structure	**Pauli interpretation**	**Result**
actual error labels $e \in V$	point masses at finite-support errors	Proposition 5.4
all pure probes	point masses at arbitrary-support labels in $\hat{V}$	Proposition 5.4
all probes	Pauli channels: probability laws on $\hat{V}$	Proposition 5.4
response of a mixture	anticommutation probability for independent draws of a check and an error	Proposition 5.4
uniform response on the 1-coset minus that on the 0-coset	probability of that specified nonzero syndrome	Proposition 7.1
commuting signed checks	operator analogue: projector onto a syndrome eigenspace	Proposition 7.1
TP_2 parameters	actual finite mixtures of checks	Theorem 7.2
path realizations	finite path: an error in V; whole path: a pure probe in $\hat{V}$	Theorem 7.2

TABLE 2. From response functions to syndrome probabilities. The label theory is simple and unstable; the theory of $\mathcal{S}_\omega$ has TP_2. Actual mixtures parameterize the array; whole paths have pure-probe realizations already in $\mathcal{S}_\omega$, whose error labels may have infinite support.

if $\mathcal{G}$ has $n-k$ independent generators, for an integer k with $0 \le k \le n$, then the code space has dimension 2^k and encodes k qubits. In the labels of Section 3, the group $\mathcal{G}$ corresponds, up to signs, to an isotropic subspace $S \subseteq \mathbb{F}_2^{2n}$, that is, a subspace on which the symplectic form ω vanishes. An error is a Pauli string e; measuring independent generators $g_1, \dots, g_{n-k}$ of $\mathcal{G}$ on the corrupted state returns the *syndrome* $(\omega(g_1, e), \dots, \omega(g_{n-k}, e))$, and the decoder chooses a recovery operation from the syndrome. The syndrome need not distinguish errors that differ by a stabilizer, since they have the same action on the code space up to a phase. An error with zero syndrome is undetectable. It is harmless if it lies in $\mathcal{G}$ up to a phase, since it then acts on the code space as a scalar, and it is a *logical error* otherwise. On labels, the errors with zero syndrome form the *centralizer* $S^\perp = \{e \in \mathbb{F}_2^{2n} : \omega(s, e) = 0 \text{ for all } s \in S\}$, which contains S and has dimension $2n - (n-k) = n + k$; a set L of *logical operators* is a set of $2k$ labels with $S^\perp = S \oplus \mathrm{span}(L)$, and the values $\omega(\ell, e)$ for $\ell \in L$ record which logical operator an error of $S^\perp$ applies to the encoded qubits. If $k \ge 1$, the *distance* of the code is the least number of qubits on which a logical error acts. A *Clifford unitary* is a unitary operator U that maps Pauli strings to Pauli strings under conjugation, up to a phase; on labels, conjugation by U is a linear map of $\mathbb{F}_2^{2n}$ that preserves ω.

Example 4.1. The five-qubit code has the stabilizer generators

$$g_1 = XZZXI, \quad g_2 = IXZZX, \quad g_3 = XIXZZ, \quad g_4 = ZXIXZ,$$

where each word stands for the tensor product of its letters (thus $XZZXI$ stands for $X \otimes Z \otimes Z \otimes X \otimes I$), the letters being the Pauli matrices X, Z and $Y = \mathrm{i}XZ$ of Section 3 and the identity I of $\mathbb{C}^2$. The code encodes one qubit, with logical operators $\bar{X} = XXXXX$ and $\bar{Z} = ZZZZZ$. Each of the fifteen errors that act as X, Y or Z on a single qubit has its own nonzero syndrome, and there are exactly fifteen nonzero syndromes of four bits; for instance, the error XIIII has syndrome $(0,0,0,1)$ and the error $IIZII$ has syndrome $(0,0,1,0)$. The code therefore corrects every error on one qubit, and its distance is at least

3; after Proposition 4.7 we exhibit a logical error that acts on exactly three qubits, which shows that the distance is exactly 3.

4.2. **Fraïssé limits and Clifford circuits.** Finite Pauli configurations carry both linear relations among their labels and the alternating form that records commutation. In Proposition 4.2 we realize every finite alternating commutation matrix on linearly independent labels. In Theorem 4.3 we extend an isometry between finite-dimensional label subspaces to a Clifford conjugation on finitely many qubits; the isometry preserves addition and the form, rather than only the commutation graph. For isotropic subspaces, whose labels commute pairwise, the signs of chosen generators can also be matched, yielding the encoding statement of Corollary 4.4. Here a Clifford circuit is a product of elementary Clifford unitaries on one or two qubits; every finite-qubit Clifford unitary admits such a decomposition up to a phase [Got97, DDM03].

Proposition 4.2. *Let d be a positive integer. If Θ is a symmetric $d \times d$ matrix over $\mathbb{F}_2$ with zero diagonal, then there are Pauli strings $P_1, \dots, P_d$ with linearly independent labels, on at most d qubits, such that, for all $i, j \in \{1, \dots, d\}$, the strings P_i and P_j anticommute exactly when $\Theta_{ij} = 1$.*

Proof. The matrix Θ defines an alternating form on $\mathbb{F}_2^d$. Let W_0 be its radical and choose a complement W of W_0; the form is nondegenerate on W, which therefore has a symplectic basis of r pairs, where $2r = \dim W$. Send the r pairs to the labels of X and Z on the first r qubits, and a basis of W_0 to the labels of X on $\dim W_0$ further qubits. This linear map is injective and preserves the form, and the images of the standard basis vectors of $\mathbb{F}_2^d$ are the required strings, on $r + \dim W_0 \le d$ qubits. □

Theorem 4.3. *Let n be a positive integer, let W and W' be subspaces of $\mathbb{F}_2^{2n}$, and let $f \colon W \to W'$ be a linear bijection with $\omega(f(v), f(w)) = \omega(v, w)$ for all $v, w \in W$. Then there is a Clifford unitary U on n qubits with $UP_vU^* = \pm P_{f(v)}$ for all $v \in W$, where P_v is the Hermitian Pauli string with label v (see Section 3). If, moreover, the subspace W is isotropic and the vectors $v_1, \dots, v_m$ form a basis of W, then the unitary U can be chosen with $UP_{v_i}U^* = P_{f(v_i)}$ for all $i \le m$.*

Proof. By Witt's theorem, the map f extends to a linear bijection τ of $\mathbb{F}_2^{2n}$ that preserves ω. Every such τ is induced by conjugation by a Clifford unitary [CRSS98, DDM03]; let U be one. Since the operator UP_vU^* is a Pauli string up to a phase, Hermitian and with square I, the phase is a sign. For the second statement, let $\varepsilon_i \in \{\pm 1\}$ be the sign with $UP_{v_i}U^* = \varepsilon_i P_{f(v_i)}$. The vectors $f(v_i)$ are linearly independent, so by nondegeneracy there is $b \in \mathbb{F}_2^{2n}$ with $(-1)^{\omega(f(v_i), b)} = \varepsilon_i$ for all i. Conjugation by P_b multiplies $P_{f(v_i)}$ by $(-1)^{\omega(f(v_i), b)}$, so the Clifford unitary P_bU has the required property. □

Corollary 4.4. *Let n be a positive integer. Any two stabilizer groups on n qubits with the same number of independent generators are conjugate by a Clifford unitary. In particular, for every integer k with $0 \le k \le n$, every stabilizer code that encodes k qubits into n has an encoding circuit made of Clifford gates: a Clifford unitary that maps the code space of the trivial code, with generators $Z_1, \dots, Z_{n-k}$, where Z_i is the phase flip on the ith qubit, onto the code space of the given code.*

Proof. Let $\mathcal{G}$ and $\mathcal{G}'$ be the two stabilizer groups, let m be their common number of independent generators, and let $g_1, \dots, g_m$ and $g'_1, \dots, g'_m$ be independent generators of $\mathcal{G}$ and of $\mathcal{G}'$, respectively, with labels v_i and v'_i for $i = 1, \dots, m$. Both families span isotropic subspaces,

so the linear map $v_i \mapsto v_i'$ preserves ω, and Theorem 4.3 gives a Clifford unitary U_0 with $U_0 P_{v_i} U_0^* = P_{v_i'}$ for all i. The square of each generator is a scalar, I or $-I$, that lies in its group, and it is therefore I, since a stabilizer group does not contain $-I$; hence $g_i = \varepsilon_i P_{v_i}$ and $g_i' = \varepsilon_i' P_{v_i'}$ for signs $\varepsilon_i, \varepsilon_i' \in \{\pm 1\}$. As in the proof of Theorem 4.3, there is $b \in \mathbb{F}_2^{2n}$ with $(-1)^{\omega(v_i', b)} = \varepsilon_i \varepsilon_i'$ for all i, and the Clifford unitary $U = P_b U_0$ satisfies $U g_i U^* = g_i'$ for all i, whence $U \mathcal{G} U^* = \mathcal{G}'$. □

On countably many qubits, every finite-dimensional subspace of V lies in the labels of the first n qubits for some n. Applying Theorem 4.3 there, and extending by the identity on the remaining qubits, shows that every isometry between finite-dimensional subspaces of (V, ω) extends to an automorphism of (V, ω) implemented by a Clifford circuit on finitely many qubits.

For the physicist or operator theorist. A Fraïssé limit is a countable structure whose finite configurations are, up to isomorphism, exactly those of a given kind and in which every symmetry between two finite configurations extends to a symmetry of the whole structure; any two such structures are isomorphic. By Proposition 4.2, every finite pattern of commutation relations occurs among Pauli strings, and by Theorem 4.3, every symmetry between finite pieces extends to a global one, implemented by a Clifford circuit. The Pauli group on countably many qubits is therefore the Fraïssé limit of the finite alternating spaces over $\mathbb{F}_2$, that is, of finite commutation patterns together with the linear relations among the labels, and Clifford circuits on finitely many qubits implement the symmetries that witness its homogeneity. The existence of encoding circuits for stabilizer codes, usually proved by an explicit construction, is the shadow of this homogeneity: by Corollary 4.4, the five-qubit code is the image of the trivial code with generators Z_1, Z_2, Z_3, Z_4, the phase flips on the first four qubits, under a Clifford circuit, because both groups have four independent generators, all commuting, and the two generating sets therefore have the same commutation pattern.

For the logician. The structure (V, ω) of Section 3 is the Pauli group modulo phases, and, for each positive integer n, the Clifford unitaries on the first n qubits act on the finite piece $(\mathbb{F}_2^{2n}, \omega)$ through its automorphism group, the symplectic group. Ultrahomogeneity, which in Section 3 came from Witt's theorem and a back-and-forth argument, is implemented by Clifford circuits. On labels, a stabilizer code becomes an isotropic subspace, which determines it up to the signs of its generators, and Corollary 4.4 follows from ultrahomogeneity for isotropic subspaces, with the signs of the generators as additional data handled by Lemma 3.2.

4.3. **Forking and logical errors.** In a simple theory, an extension of the type of an element forks when, in the sense recalled in Section 2 and explained after Proposition 4.7, it confines the element to one of infinitely many alternatives that the original type cannot tell apart. The next theorem computes forking for the Pauli group, using the description of nonforking independence in Theorem 3.3.

Theorem 4.5. *Let e be an element of a monster model $\mathbb{M}$ of the theory of (V, ω), and let $D \subseteq B$ be small subsets of $\mathbb{M}$. Then the type $\mathrm{tp}(e/B)$ forks over D if and only if $e \in \mathrm{span}(B) \setminus \mathrm{span}(D)$.*

Proof. By Theorem 3.3, the type $\mathrm{tp}(e/B)$ does not fork over D exactly when $\mathrm{span}(eD) \cap \mathrm{span}(B) = \mathrm{span}(D)$. If $e \in \mathrm{span}(D)$, the spans $\mathrm{span}(eD)$ and $\mathrm{span}(D)$ coincide, and the equality holds. Otherwise the span of eD is $\mathrm{span}(D) \oplus \{0, e\}$, and the equality fails if and only if some vector $e + d$ with $d \in \mathrm{span}(D)$ lies in $\mathrm{span}(B)$, that is, if and only if $e \in \mathrm{span}(B)$. □

By Theorem 4.5, commutation data never fork: the type of an element e of $\mathbb{M}$ over a small set B records the values $\omega(b, e)$ for $b \in B$, and these values, constrained only by linearity and by the type of e over a subset D of B, can be prescribed without forking over D. Forking happens only when e becomes a linear combination of elements of B; for Pauli strings, this means that e becomes, up to a phase, a product of strings in B. In a stabilizer code this criterion separates the two kinds of nonzero undetectable error. To state the consequence for the distance of a code, we call the number of qubits on which an error acts nontrivially its *weight*.

Corollary 4.6. *Let n be a positive integer, and regard $\mathbb{F}_2^{2n}$ as the subspace of V formed by the labels of the strings on the first n qubits. Let $\mathcal{G}$ be a stabilizer group on n qubits with set of labels S, and let $e \in \mathbb{F}_2^{2n}$ be a nonzero label with zero syndrome. Then the type $\mathrm{tp}(e/S)$ forks over $\emptyset$ if and only if e is harmless, that is, lies in S. Consequently, if the code of $\mathcal{G}$ encodes at least one qubit, its distance is the least weight of a nonzero label in $\mathbb{F}_2^{2n}$ with zero syndrome whose type over S does not fork over $\emptyset$.*

Proof. Apply Theorem 4.5 with $D = \emptyset$ and $B = S$; the span of S is S itself, and $\mathrm{span}(\emptyset) = \{0\}$. The second statement is the definition of distance, since a logical error is a nonzero label with zero syndrome outside S. □

In Proposition 4.7 below, we show that the stabilizer group alone does not tell logical errors apart, because they all have the same type over its set of labels, and that the logical operators, once known, characterize them by forking.

Proposition 4.7. *Let n be a positive integer, regard $\mathbb{F}_2^{2n}$ as a subspace of V as in Corollary 4.6, let $\mathcal{G}$ be a stabilizer group on n qubits with set of labels S, let L be a set of logical operators of its code, and let $e \in \mathbb{F}_2^{2n}$ be a label with zero syndrome. Then the following hold:*

(a) *the label e is a logical error if and only if $\mathrm{tp}(e/S \cup L)$ forks over S;*
(b) *all logical errors have the same type over S, and this type does not fork over $\emptyset$.*

Proof. (a) Since the syndrome of e is zero, the label e lies in $S^\perp = \mathrm{span}(S \cup L)$. By Theorem 4.5 with $D = S$ and $B = S \cup L$, the type $\mathrm{tp}(e/S \cup L)$ forks over S if and only if $e \in \mathrm{span}(S \cup L) \setminus S = S^\perp \setminus S$, which is the set of logical errors.

(b) Suppose that the label e is a logical error, that is, $e \in S^\perp \setminus S$. By quantifier elimination, the type of e over S is determined by the linear relations of e over S, of which there are none since $e \notin S$, and by the values of ω on $\mathrm{span}(eS)$. These values are fixed, through bilinearity, by the values of ω on S, by $\omega(e, e)$, which is 0 because ω is alternating, and by the values $\omega(s, e)$ for $s \in S$, which are 0 because e lies in $S^\perp$. Hence all logical errors have the same type over S. The label e is nonzero, since the zero label lies in S; as it has zero syndrome and lies outside S, its type over S does not fork over $\emptyset$, by Corollary 4.6. □

In the five-qubit code, the errors $g_1 g_2 = XYIYX$ (up to a phase) and $\bar{X} = XXXXX$ both have zero syndrome. The first lies in the stabilizer group, whose set of labels we denote by S, so its type over S forks over $\emptyset$; it is harmless. The second does not, so its type does not fork over $\emptyset$; it flips the encoded qubit, and it acts on five qubits. By Example 4.1, an error of the second kind acts on at least three qubits, and the logical error $\bar{X} g_1 = IYYIX$, up to a phase, acts on exactly three; the distance of the code is therefore 3. With $L = \{\bar{X}, \bar{Z}\}$, the logical errors $\bar{X}$, $\bar{Z}$ and $\bar{X} g_1$ have the same type over S; over $S \cup L$ each of them lies in $\mathrm{span}(S \cup L)$, whence its type becomes algebraic and forks over S, as Proposition 4.7(a) predicts.

For the physicist or operator theorist. Forking is the model-theoretic notion of an observation that singles out one of infinitely many alternatives that the data known in advance cannot tell apart and of which, for some fixed integer $k \geq 2$, no k can hold together. To be precise, in a simple theory, by a theorem of Kim [Kim98], a condition $\varphi(x, b)$ on an unknown element x forks over a set D of known parameters exactly when, for some $k \geq 2$, the parameter b is the first term of a sequence $(b_j)_{j\in\mathbb{N}}$ that is indiscernible over D and k-inconsistent for φ, in the sense that no k of the conditions $\varphi(x, b_j)$ can hold together (see Definition 2.1). A sequence is indiscernible over D if any two of its increasing finite subsequences of the same length satisfy the same formulas with parameters from D. A type forks over D when one of its conditions does; such a type narrows the possibilities for the element but need not determine it. For a single Pauli label e, Theorem 4.5 gives the exact criterion: if $D \subseteq B$ are parameter sets, then $\mathrm{tp}(e/B)$ forks over D precisely when $e \in \mathrm{span}(B) \setminus \mathrm{span}(D)$. Thus a forking extension fixes the label as a sum of the added parameters, although a complete commutation pattern need not fix a label. For the stabilizer label space S of a finite code and a nonzero error label e with zero syndrome, the base matters. Over $\emptyset$, the type $\mathrm{tp}(e/S)$ forks exactly when $e \in S$, the harmless case; a logical error lies in $S^\perp \setminus S$ and its type over S does not fork over $\emptyset$ (see Corollary 4.6). For a code encoding at least one qubit, its distance is the least weight of such a logical error on the code's qubits. Now add logical labels L with $S^\perp = S \oplus \mathrm{span}(L)$. The same logical error has $\mathrm{tp}(e/S \cup L)$ forking over S, because it is in the enlarged span but not in S (see Proposition 4.7). Without these added labels, all logical errors have the same type over S; nonforking extensions to $S \cup L$ remain possible outside $\mathrm{span}(S \cup L)$, for example on qubits beyond the code, as the next reading explains. These are statements about dependence and available parameter sets, not a rule for inferring a unique error from its measured syndrome.

For the logician. The theory of (V, ω) has a vector space pregeometry for algebraic closure, and by Theorem 4.5 its forking is the dependence of this pregeometry: the type of a single element over a set forks over a subset exactly when the element is algebraic over the set but not over the subset. Thus the theory has SU-rank one, as does the theory of the Rado graph (see Theorem 3.1), with linear span in place of equality. In finite dimension, the information carried by a syndrome is measured by counting: if n and m are positive integers and $s_1, \ldots, s_m$ are linearly independent labels in $\mathbb{F}_2^{2n}$, then for each syndrome $\sigma \in \mathbb{F}_2^m$ exactly 2^{2n-m} labels e satisfy $\omega(s_i, e) = \sigma_i$ for all i, since the map $e \mapsto (\omega(s_i, e))_{i\leq m}$ is linear and onto. This count records codimension, which forking ignores. With S the set of labels of the stabilizer group of a code, the logical errors of the code are the elements of the centralizer of S outside S, and by Proposition 4.7 they share one type over S. For a set L of logical operators, the nonforking extensions of this type to $S \cup L$ are realized in (V, ω) by labels outside $\mathrm{span}(S \cup L)$ that commute with S, for instance by errors on qubits that the code does not touch; the logical errors realize the forking extensions of the same type, which become algebraic once the logical operators are added.

4.4. **The independence theorem and the gluing of syndromes.** The Kim–Pillay independence theorem combines two nonforking extensions of one type over a model when their parameter sets are independent over that model. For Pauli labels, the corresponding compatibility condition is linear: the spans of the two sets of checks meet only in the span of their common base, and the two syndrome functionals agree there. The proposition below states this finite-dimensional counterpart over the span of an arbitrary common set, rather than assuming that the base is a model. It asserts existence of a label with both prescribed syndromes; it imposes neither a weight bound nor a choice of decoder. Pairwise commutation

is needed to interpret a set of checks as simultaneous stabilizer measurements, but not for the linear compatibility statement itself.

Proposition 4.8. *Let n be a positive integer, let C, B and D be subsets of $\mathbb{F}_2^{2n}$ with* $\operatorname{span}(BC) \cap \operatorname{span}(DC) = \operatorname{span}(C)$*, and let ξ_1 and ξ_2 be linear functionals on* $\operatorname{span}(BC)$ *and on* $\operatorname{span}(DC)$*, respectively, that agree on* $\operatorname{span}(C)$*. Then there is a label $e \in \mathbb{F}_2^{2n}$ with $\omega(s, e) = \xi_1(s)$ for $s \in \operatorname{span}(BC)$ and $\omega(s, e) = \xi_2(s)$ for $s \in \operatorname{span}(DC)$.*

Proof. Since ξ_1 and ξ_2 agree on the intersection of their domains, there is a linear functional ξ on $\operatorname{span}(BC) + \operatorname{span}(DC)$ extending both. Extend ξ to $\mathbb{F}_2^{2n}$. Because ω is nondegenerate, every linear functional on $\mathbb{F}_2^{2n}$ has the form $\omega(\cdot, e)$ for a unique e. □

In the five-qubit code, take $C = \emptyset$, $B = \{g_1, g_2\}$ and $D = \{g_3, g_4\}$. The spans of B and D meet only in 0, so any two bits of syndrome on g_1, g_2 and any two on g_3, g_4 are realized together by one error; by Example 4.1, every nonzero syndrome is realized even by an error on a single qubit, for instance the syndrome $(1, 0 \,|\, 1, 1)$ by $YIIII$. If D is replaced by $\{g_1 g_2\}$, the hypothesis fails, and so does the conclusion: the syndrome bit of every error with respect to $g_1 g_2$ is the sum of its bits with respect to g_1 and g_2.

For the physicist or operator theorist. For finite-support Pauli labels, compatible syndrome data are linear functionals on the spans of the checks. If these spans intersect only in the common base span, agreement there gives a joint syndrome, as Proposition 4.8 states. The five-qubit example also shows why the overlap must include linear relations: a prescribed bit on $g_1 g_2$ must equal the sum of the prescribed bits on g_1 and g_2. Together with the other axioms of Fact 2.2, the model-theoretic independence theorem characterizes simplicity [KP97]. The passage to the mixed-preparation observable changes the structure and its theory; it does not invalidate this linear gluing statement. In Section 6 we construct the tree property of the second kind for response formulas in the real-valued structure of Alice's mixtures and Bob's probes. For the Pauli observable, Theorem 7.2 supplies an array of actual finitely supported mixture parameters. Every finite selection along a path is realized by a finite-support error, while the entire path is realized by a pure probe in the compactified space, whose error label may have infinite support; this pure probe already belongs to the original observable structure. The incompatibility within each row holds for every probe, including mixed ones. Thus the simple unstable theory of the symplectic labels and the nonsimple theory of the mixed response structure have different mathematical domains, even though both use the same commutation pairing.

For the logician. The statement of Proposition 4.8 is the counterpart, in finite dimension and over the span of an arbitrary set rather than over a model, of the independence theorem verified in the proof of Theorem 3.3: the whole argument is the gluing of two functionals whose domains, the spans of two sets of checks together with a common set, meet only in the span of the common set. Its physical content is that the syndrome bits of an error with respect to a set of checks are the values of a linear functional on the span of the checks, and the consistency of prescribed syndromes is therefore a matter of linear algebra on that span; the surjectivity of the syndrome map of linearly independent checks is Lemma 3.2.

5. Observables, mixtures and dividing

We now pass from a relation on labels to an observable and its mixed preparations. This changes the structure whose theory is being studied: a probability distribution on labels becomes one preparation, and the response records its average against a probe. Although the

Pauli and Rado label theories are both simple and unstable, their observables have different probe spaces: linear functionals on the Pauli space and arbitrary sets of Rado vertices describe the pure probes. We construct the two response structures and formulate the tree properties for their real-valued predicates, preparing the comparison in the following sections.

5.1. **Diagonal observables.** Let A and B be countable sets and let $\varphi\colon A \times B \to \{0,1\}$ be a function. The *diagonal observable* of φ is the operator

$$C_\varphi = \sum_{a\in A,\, b\in B} \varphi(a,b)\, |a\rangle\langle a| \otimes |b\rangle\langle b| \quad \text{on } \ell^2(A)\otimes\ell^2(B).$$

Here $|a\rangle\langle a|$ and $|b\rangle\langle b|$ are the orthogonal projections onto the lines spanned by the basis vectors of $\ell^2(A)$ and $\ell^2(B)$ indexed by a and b, respectively. In plain words, Alice's register holds a label a, Bob's a label b, and the observable reads $\varphi(a,b)$. A density operator ρ of Alice, that is, a positive operator of trace one on $\ell^2(A)$ that describes her preparation, enters only through its diagonal, which is a probability measure μ on A, a *mixture* of her labels: the preparation draws a at random with law μ. The operator on Bob's space obtained by averaging C_φ over Alice's preparation, the effective observable of ρ in the terminology of the second paper, is the diagonal operator of multiplication by

$$b \longmapsto \sum_{a\in A} \mu(a)\varphi(a,b).$$

The probe algebra of the second paper is the unital C*-algebra generated by these effective observables, that is, the smallest algebra of operators on Bob's space that contains them and the identity and is closed under adjoints and under limits in the operator norm; it is the commutative C*-algebra of bounded functions on B generated by the constants and the indicators $b \mapsto \varphi(a,b)$, $a \in A$. Bob's probes are the states of the probe algebra, the positive linear functionals on it with value 1 at the identity, and they form the probe space, which carries the weak* topology, the topology of pointwise convergence on the probe algebra. The probe space is described through the closure K_φ of the set of functions $\varphi(\cdot,b)$, $b \in B$, in $\{0,1\}^A$ with the product topology.

Lemma 5.1. *Let A and B be countable sets and let $\varphi\colon A\times B \to \{0,1\}$ be a function. Then the probe algebra of the diagonal observable C_φ is isomorphic to the algebra $C(K_\varphi)$ of continuous complex functions on the closure K_φ of the set of functions $\varphi(\cdot,b)$, $b \in B$, in $\{0,1\}^A$, the probe space is the space $\mathcal{P}(K_\varphi)$ of Radon probability measures on K_φ with the weak* topology, and the response of a mixture μ, a probability measure on A, is the affine continuous function*

$$r_\mu(\nu) = \int_{K_\varphi} \sum_{a\in A} \mu(a)\lambda(a)\, d\nu(\lambda) \qquad (\nu \in \mathcal{P}(K_\varphi)).$$

Proof. For each $a \in A$, let q_a be the coordinate function $q_a(\lambda) = \lambda(a)$ on K_φ. These functions are real-valued and separate points of K_φ. The unital complex algebra they generate is therefore closed under complex conjugation and uniformly dense in $C(K_\varphi)$ by the Stone–Weierstrass theorem. Restriction along $b \mapsto \varphi(\cdot,b)$ preserves the supremum norm, because this map has dense image in K_φ, and sends q_a to the indicator $b \mapsto \varphi(a,b)$. It thus identifies $C(K_\varphi)$ with the probe algebra.

By the Riesz representation theorem, a state of $C(K_\varphi)$ is integration against a unique Radon probability measure; pointwise convergence on $C(K_\varphi)$ is precisely the stated weak* topology on these measures. For a mixture μ, the series $\sum_a \mu(a)q_a$ converges uniformly, since

the norm of a tail is bounded by the corresponding probability mass. This function corresponds to the effective observable of the preparation with diagonal μ. Integrating it against the measure ν gives the displayed response formula, which is affine and weak* continuous in ν. □

The pure probes are the point masses δ_λ, $\lambda \in K_\varphi$, and we identify each of them with its point λ; a pure probe is thus an *idealized label* of Bob, recording which of Alice's labels it answers with 1. The actual labels, the functions $\varphi(\cdot, b)$ for $b \in B$, are dense among them, since K_φ is the closure of the set of these functions. For a label b, we write $r_\mu(b)$ for the response at the pure probe $\varphi(\cdot, b)$.

5.2. **The structure of an observable.** The theory of the observable concerns its mixtures and probes, rather than its label structure alone. For a diagonal observable C_φ, let $\mathcal{S}_\varphi$ be the two-sorted structure whose preparation sort consists of Alice's mixtures, the probability measures on A, whose probe sort is $\mathcal{P}(K_\varphi)$, and whose only predicate is the response $R(\mu, \nu) = r_\mu(\nu)$, with values in $[0, 1]$. As in the second paper, we use Keisler's real-valued logic [Kei23], in which no metric is prescribed in advance; in an ultrapower, elements of the same sort are identified when all atomic formulas with parameters have the same values at them. We call C_φ *simple* when the real-valued theory of $\mathcal{S}_\varphi$ is simple in the sense of Ben Yaacov [BY03]. An elementary extension is a larger structure in which every formula with parameters from $\mathcal{S}_\varphi$ retains its value; its new elements of the preparation sort are called idealized preparations. For the Pauli observable, the arbitrary-support error labels described in the Introduction already occur as pure probes in the original structure, as we show in Proposition 5.4; elementary extension is a further enlargement. Every tree property of the theory of $\mathcal{S}_\varphi$ is witnessed in an $\aleph_1$-saturated elementary extension, one realizing every countable finitely satisfiable set of conditions with parameters. We say that actual preparations witness the property when its preparation parameters lie in $\mathcal{S}_\varphi$ itself and its finite path fragments are realized in the original probe sort, in the sense of Definition 5.2. For the Pauli observable, this does not require a whole infinite path to be realized by an error of finite support.

5.3. **Dividing for families of functions.** The passage from truth values to expectations expresses inconsistency through uniformly separated real values. As in Part II, an independent sequence of responses realizes every finite high/low pattern at two fixed thresholds. For the tree properties, individual conditions are satisfiable, but fixed-size subfamilies within rows or sibling sets are uniformly inconsistent, while finite portions of each path remain jointly satisfiable. The common positive gap supplies the inequalities required in the real-valued formulation, and these finite configurations persist in elementary extensions of the response structure $\mathcal{S}_\varphi$. We state the function-family formulation before constructing the arrays of mixtures.

Definition 5.2. Let Y be a set and $\mathcal{R}$ a family of real functions on Y.

(1) For an integer $k \geq 2$, a sequence $(r_j)_{j\in\mathbb{N}}$ in $\mathcal{R}$ is *uniformly k-inconsistent* if there are reals $\alpha < \beta$ such that each r_j takes a value $\geq \beta$ somewhere on Y, and for all $j_1 < \cdots < j_k$ no $y \in Y$ satisfies $r_{j_\ell}(y) > \alpha$ for $\ell = 1, \ldots, k$.

(2) The family $\mathcal{R}$ has the *tree property of the second kind* (TP_2) if there are $k \geq 2$, reals $\alpha < \beta$ and an array $(r_{ij})_{i,j\in\mathbb{N}}$ in $\mathcal{R}$ such that every row is uniformly k-inconsistent with the constants α and β, and for every $f\colon \mathbb{N} \to \mathbb{N}$ and every $n \in \mathbb{N}$ there is $y \in Y$ with $r_{if(i)}(y) \geq \beta$ for all $i < n$.

The tree property of Definition 2.1, its variant of the first kind, and TP_2 all require infinite sequences of instances that are k-inconsistent, for some integer $k \geq 2$: the children of a node,

or a row. When these notions are transported to a family of functions as in Definition 5.2, those sequences become uniformly k-inconsistent. A family that, for no integer $k \geq 2$, contains a uniformly k-inconsistent sequence therefore has none of the three tree properties. Applied to the family of responses of an observable, condition (2) of Definition 5.2 says that actual preparations witness the tree property of the second kind of the atomic formula, in the real-valued logic of the second paper; the gap between the constants α and β of Definition 5.2 is what the definitions of dividing in real-valued logic require [BY03].

5.4. **The Pauli observable.** The *Pauli observable* is the diagonal observable of the symplectic form ω on $V \times V$, where V is the space of labels of the Pauli strings on countably many qubits that act nontrivially on only finitely many of them (see Section 3): Alice and Bob each hold a Pauli label, and the observable reports whether the two Pauli strings anticommute. We call Alice's label the check and Bob's label the error; the observable then reports one syndrome bit. For a mixture μ, the response at an error e is the probability that a check drawn with law μ anticommutes with e. By Lemma 3.2, which provides an error with any prescribed syndrome with respect to finitely many linearly independent checks, the observable has the independence property already on pure preparations.

Lemma 5.3. *For the Pauli observable, the space K_ω is the group V^* of all linear functionals $V \to \mathbb{F}_2$.*

Proof. Each $\omega(\cdot, e)$ is linear, and the linear functionals form a closed subset of $\{0,1\}^V$. Conversely, a basic neighborhood of a linear functional λ prescribes the values of λ on finitely many vectors, which may be taken linearly independent, and Lemma 3.2 gives e with $\omega(\cdot, e)$ in the neighborhood. □

The functionals of V^* have a direct physical meaning. Let $\hat{V} = \prod_{k\in\mathbb{N}} \mathbb{F}_2^2$, with the product topology, be the group of labels of Pauli strings of arbitrary support, the support of a string being the set of qubits on which it acts nontrivially; it contains V as the strings of finite support. For $a \in V$ and $u \in \hat{V}$, the value $\omega(a, u)$ is defined by the formula of Section 3, which is a finite sum because a has finite support. For p with $0 < p \leq 1$, let ν_p be the *depolarizing noise* of rate p: the product measure on $\hat{V}$ under which the labels on distinct qubits are independent, each equal to $(0,0)$ with probability $1-p$ and to each of $(1,0)$, $(0,1)$ and $(1,1)$ with probability $p/3$. A Radon probability measure on $\hat{V}$ specifies a *Pauli channel* on the norm closure of the algebra of operators supported on finitely many qubits: on each finite set of qubits, the channel averages conjugation by the Pauli strings given by the restrictions of the random label to that set. We identify the channel with this probability law. An arbitrary-support label thus specifies compatible actions on all finite sets of qubits; no infinite tensor product unitary on a chosen Hilbert space is required.

Proposition 5.4. *Let $\hat{V}$ be the group of labels of Pauli strings of arbitrary support, and for $0 < p \leq 1$ let ν_p be the depolarizing noise of rate p. Then the following hold:*

(a) *the map $u \mapsto \omega(\cdot, u)$ is an isomorphism of topological groups from $\hat{V}$ onto V^*, which carries V onto the functionals of the actual errors;*
(b) *the probes of the Pauli observable are the Pauli channels on countably many qubits, and the response of a mixture μ at a Pauli channel ν is the probability that a check drawn with law μ anticommutes with an independent error drawn with law ν;*
(c) *the depolarizing noise ν_p is concentrated on errors of infinite support, and for a label $a \in V$ that acts nontrivially on w qubits we have*

$$r_{\delta_a}(\nu_p) = \tfrac{1}{2}\big(1 - (1 - \tfrac{4p}{3})^w\big).$$

Proof. (a) The map is a homomorphism. Write X_k and Z_k for the labels of the strings X and Z on the qubit k. Then the values $\omega(X_k, u)$ and $\omega(Z_k, u)$ are the two coordinates of u at the qubit k, so the map is injective, and every $\lambda \in V^*$ is the image of the vector $u = (x, z)$ with $z_k = \lambda(X_k)$ and $x_k = \lambda(Z_k)$ for every k, because λ and $\omega(\cdot, u)$ are linear and agree on the basis $\{X_k, Z_k : k \in \mathbb{N}\}$ of V. The value $\omega(a, u)$ depends on finitely many coordinates of u, so the map is continuous, and a continuous bijection between compact Hausdorff spaces is a homeomorphism.

(b) By Lemma 5.1, Lemma 5.3 and (a), the probes are the Radon probability measures on $\hat{V}$, and $r_\mu(\nu) = \int \sum_a \mu(a)\,\omega(a, u)\,d\nu(u)$, which is the probability stated, by Fubini's theorem.

(c) Under ν_p, the events "the error acts nontrivially on the qubit k" are independent and have probability p, so, by the Borel–Cantelli lemma, almost every error acts nontrivially on infinitely many qubits. At each of the w qubits where the string with label a acts nontrivially, its factor anticommutes with two of the three nontrivial Pauli matrices, hence with the factor of the error with probability $2p/3$, independently of the other qubits. The strings anticommute exactly when an odd number of these w events occur, and this has probability $\frac{1}{2}\big(1 - (1 - \frac{4p}{3})^w\big)$. □

In what follows we identify V^* with $\hat{V}$ through Proposition 5.4(a) and call its elements *idealized errors*. By Proposition 5.4(c), the depolarizing noise ν_p of each rate p, which models the noise of a memory on countably many qubits, is a probe of the Pauli observable concentrated on idealized errors that are not actual: with probability one, an error drawn from ν_p acts on infinitely many qubits. Alice's checks see such an error through finitely many qubits at a time, and the response of a check of weight w, one that acts nontrivially on w qubits, tends to $\frac{1}{2}$, the value of a fair coin, as w grows. For $a \in V$, let χ_a be the character of V^* given by $\chi_a(\lambda) = (-1)^{\lambda(a)}$, a continuous homomorphism of V^* into the multiplicative group $\{\pm 1\}$. Alice's pure responses are the characters of this compact group up to an affine change: the pure response r_{δ_a} of a label a takes the value $\lambda(a) = (1 - \chi_a(\lambda))/2$ at an idealized error λ.

5.5. **The Rado observable.** The *Rado observable* is the diagonal observable of the edge relation E on $\Gamma \times \Gamma$, where (Γ, E) is the Rado graph of Section 3: Alice and Bob each hold a site of a random network of couplings, and the observable reports whether the two sites are coupled. By the extension property of the Rado graph, which provides a vertex adjacent to every vertex of a finite set and to none of another finite set disjoint from it, the observable has the independence property already on pure preparations. We say that the diagonal observable C_φ of a function $\varphi\colon A \times B \to \{0, 1\}$ has *freely independent* pure responses when $K_\varphi = \{0, 1\}^A$, that is, when every pattern of answers to all of Alice's labels is an idealized label of Bob.

Lemma 5.5. *For the Rado observable, the space K_E is the whole Cantor cube $\{0, 1\}^\Gamma$. Hence the Rado observable has freely independent pure responses, while the Pauli observable does not.*

Proof. A basic open subset of $\{0, 1\}^\Gamma$ prescribes adjacency to the vertices of a finite set Γ_1 and nonadjacency to those of a finite set Γ_0, and the extension property gives a vertex outside $\Gamma_1 \cup \Gamma_0$ with that pattern. By Lemma 5.3, the space K_ω is the group V^*, which is a proper subset of $\{0, 1\}^V$. □

The pure probes of the Rado observable are arbitrary sets of neighbors, that is, arbitrary subsets of Γ, and its probes are random sets of neighbors. The response of a mixture μ at a probe ν is the probability that a site drawn with law μ lies in a set of neighbors drawn, independently, with law ν; it depends on ν only through the marginals $\nu\{\lambda : \lambda(a) = 1\}$, for

$a \in \Gamma$, the probabilities that the individual sites are neighbors. The Pauli observable sees more of a probe: its pure responses are, up to an affine change, the characters of the compact group of idealized errors, and products of characters are again characters. In Sections 7 and 8 we show what this difference does to the tree property.

6. The theory of mixed preparations has the tree property of the second kind

Passing from labels to mixtures changes the structure whose theory is being considered. For a diagonal observable with the independence property, we show that the real-valued theory of preparations and probes has the tree property of the second kind, witnessed by the difference of two responses. The construction compares the uniform mixture on a half of a finite set of labels with the uniform mixture on its complement. Their difference is 1 precisely when the probe answers 1 on the first half and 0 on the second; two halves meeting in a quarter of the set cannot both give a difference greater than $1/2$. The independence property supplies compatible answers on disjoint blocks, yielding finite arrays of actual preparation parameters of arbitrarily large size. Compactness then supplies the infinite array in an elementary extension (see Theorem 6.2); whether its parameters can all be actual preparations depends on the observable.

The construction uses the independence property on actual labels, whereas NIP for the observable concerns all mixed responses on the full probe space. We first separate the analytic fact that identifies these two conditions.

Lemma 6.1 (NIP for pure and mixed responses). *Let A and B be countable sets and let C_φ be the diagonal observable of a function $\varphi\colon A \times B \to \{0,1\}$. Recall from Lemma 5.1 that K_φ is the closure of $\{\varphi(\cdot,b) : b \in B\}$ in $\{0,1\}^A$, and the full probe space is $\mathcal{P}(K_\varphi)$. For $a \in A$, put $f_a(\lambda) = \lambda(a)$ for $\lambda \in K_\varphi$. For a probability distribution μ on A, put $g_\mu = \sum_{a\in A} \mu(a) f_a$; its response is $r_\mu(\nu) = \int g_\mu \, d\nu$ for $\nu \in \mathcal{P}(K_\varphi)$. With NIP meaning the absence of an infinite independent sequence at fixed thresholds, the following are equivalent:*

(i) *the family $\{\varphi(a,\cdot) : a \in A\}$ is NIP on B;*
(ii) *the family $\{f_a : a \in A\}$ of pure responses is NIP on K_φ;*
(iii) *the family $\{r_\mu : \mu$ is a probability distribution on $A\}$ of mixed responses is NIP on the full probe space $\mathcal{P}(K_\varphi)$.*

Proof. The functions f_a take only the values 0 and 1, so independence for them is equivalent to realizing every finite pattern of these two values. Each such pattern specifies a clopen subset of K_φ. If it is nonempty, it meets the dense set of actual labels $\{\varphi(\cdot,b) : b \in B\}$. Conversely, every pattern realized by an actual label is realized in K_φ. This proves (i)$\Leftrightarrow$(ii). If the pure responses are independent, the mixed responses associated with the point masses on the same labels are independent, by evaluation at the pure probes δ_λ. Thus (iii) implies (ii).

Assume (ii). Rosenthal's combinatorial dichotomy for uniformly bounded sequences of real functions [Ros74] gives every sequence of pure responses a subsequence that converges pointwise on K_φ, since an independent subsequence is excluded. This subsequence is weakly Cauchy in the real Banach space $C(K_\varphi)$ of continuous real-valued functions with the supremum norm: its values under each continuous linear functional form a Cauchy sequence. Indeed, every such functional is integration against a finite signed Radon measure, and the functions are bounded by 1, so dominated convergence applies. Hence $\{f_a : a \in A\}$ is weakly precompact, meaning that it is bounded and every sequence in it has a weakly Cauchy subsequence.

Stegall's theorem says that the absolutely convex hull of a weakly precompact set is weakly precompact (see [Rod21]). We use it here as in Part II [DI26b]. Here the absolutely convex hull consists of the finite sums $\sum_j c_j f_{a_j}$ with $\sum_j |c_j| \leq 1$. Its norm closure is also weakly precompact: given a sequence $(h_n)_{n\in\mathbb{N}}$ in the closure, choose functions q_n in the hull with $\|h_n - q_n\|_\infty < 2^{-n}$ for $n \in \mathbb{N}$ and a weakly Cauchy subsequence of (q_n); the corresponding subsequence of (h_n) is weakly Cauchy as well. Each g_μ belongs to this closure, because its finite partial sums lie in the hull and converge uniformly, with the omitted probability mass bounding the error.

Every sequence (g_{μ_n}) therefore has a weakly Cauchy subsequence. Since integration against each probe $\nu \in \mathcal{P}(K_\varphi)$ is a continuous linear functional on $C(K_\varphi)$, the corresponding subsequence of responses (r_{μ_n}) converges pointwise on the full probe space. An independent sequence of these continuous responses cannot have a pointwise convergent subsequence. To see this, prescribe alternating low and high values along any subsequence at its fixed thresholds. Every finite prescription is realized, and the associated closed subsets of the compact probe space have the finite intersection property; hence one probe realizes the entire alternating prescription, preventing convergence there. Thus the mixed family is NIP, proving (iii). □

We now use an independent sequence of actual labels to construct the arrays of mixtures.

Theorem 6.2 (Mixed preparations and the tree property of the second kind). *Let C_φ be the diagonal observable of a function $\varphi\colon A \times B \to \{0,1\}$ on countable sets A and B, and assume that there are distinct labels $a_0, a_1, \dots$ in A such that for all finite disjoint $N_1, N_0 \subseteq \mathbb{N}$ there is a label $b \in B$ with $\varphi(a_n, b) = 1$ for $n \in N_1$ and $\varphi(a_n, b) = 0$ for $n \in N_0$. Then the affine formula*

$$\psi(x; y_1, y_2) = R(y_1, x) - R(y_2, x),$$

in which the predicate R is the response, the variable x is of the sort of probes and the variables y_1, y_2 are of the sort of preparations, has the tree property of the second kind in the real-valued theory of the structure $\mathcal{S}_\varphi$ of Alice's mixtures and Bob's probes, with $k = 2$, $\alpha = 1/2$ and $\beta = 1$. Here k, α and β are the constants of Definition 5.2, applied to the family of instances of ψ, as functions on the probes, in an $\aleph_1$-saturated elementary extension of $\mathcal{S}_\varphi$. Moreover, every simple diagonal observable is NIP (see Lemma 6.1).

Proof. We first build finite arrays in $\mathcal{S}_\varphi$ itself and then pass to an elementary extension by compactness.

Let $N \geq 1$ be an integer, and choose an integer t with $2^t > N$. Partition $2^t N$ of the labels a_n into N blocks $J_0, \dots, J_{N-1}$ of size 2^t, and identify each block with $\mathbb{F}_2^t$. Choose distinct nonzero vectors $w_0, \dots, w_{N-1}$ in $\mathbb{F}_2^t$, and for $i, j < N$ let $H_{ij} \subseteq J_i$ be the set of $v \in \mathbb{F}_2^t$ with $w_j \cdot v = 1$, where the dot product $w_j \cdot v$ is computed in $\mathbb{F}_2$. Each H_{ij} has 2^{t-1} elements, and for $j \neq j'$ the set $H_{ij} \cap H_{ij'}$ has 2^{t-2} elements, since w_j and $w_{j'}$ are linearly independent. Let μ_{ij}^+ and μ_{ij}^- be the uniform probability measures on H_{ij} and on $J_i \setminus H_{ij}$.

For a probe ν, write $\pi_a(\nu) = \nu\{\lambda : \lambda(a) = 1\} \in [0,1]$ for its marginal at the label a; by Lemma 5.1, the response is then $R(\mu, \nu) = \sum_a \mu(a)\pi_a(\nu)$. With $m = 2^t$, we have

$$\psi(\nu; \mu_{ij}^+, \mu_{ij}^-) = \frac{2}{m}\Big(\sum_{a\in H_{ij}} \pi_a(\nu) - \sum_{a\in J_i\setminus H_{ij}} \pi_a(\nu)\Big) \leq 1.$$

For $j \neq j'$ in the same row, adding the two values gives

$$\frac{4}{m}\Big(\sum_{a\in H_{ij}\cap H_{ij'}} \pi_a(\nu) - \sum_{a\in J_i\setminus(H_{ij}\cup H_{ij'})} \pi_a(\nu)\Big) \le \frac{4}{m}\cdot\frac{m}{4} = 1,$$

so the two values are never both greater than $1/2$. For every function f from $\{0,\dots,N-1\}$ to itself, the hypothesis applied to the finite disjoint sets $\bigcup_i H_{if(i)}$ and $\bigcup_i (J_i \setminus H_{if(i)})$ gives a label b, and at the pure probe of $\varphi(\cdot, b)$ every value $\psi(\cdot; \mu^+_{if(i)}, \mu^-_{if(i)})$ with $i < N$ equals 1.

Thus, for every N, the structure $\mathcal{S}_\varphi$ satisfies the following conditions on $2N^2$ preparations $y^\pm_{ij}$, for $i, j < N$: for every function f from $\{0,\dots,N-1\}$ to itself, the value of $\sup_x \min_{i<N} \psi(x; y^+_{if(i)}, y^-_{if(i)})$ is 1; and for all $i < N$ and all distinct $j, j' < N$, the value of $\sup_x \min\big(\psi(x; y^+_{ij}, y^-_{ij}), \psi(x; y^+_{ij'}, y^-_{ij'})\big)$ is at most $1/2$. Impose these closed conditions on an infinite array of preparation variables. Every finite subset of the conditions concerns a finite subarray and is satisfied by one of the arrays just constructed, so compactness yields an array $(y^\pm_{ij})_{i,j\in\mathbb{N}}$ in an elementary extension of $\mathcal{S}_\varphi$. Pass to an $\aleph_1$-saturated elementary extension containing this array. For each $f\colon \mathbb{N}\to\mathbb{N}$, the countable set of conditions $\psi(x; y^+_{if(i)}, y^-_{if(i)}) = 1$, for $i \in \mathbb{N}$, is approximately satisfiable on every finite subset, by the supremum conditions; saturation therefore realizes all of them at one probe. Together with the row bound, this gives the tree property of the second kind, which is incompatible with simplicity in real-valued logic [BY03].

For the final assertion, suppose that the diagonal observable is not NIP. By Lemma 6.1, its pure responses on the actual labels B have an independent sequence. Since these responses take only the values 0 and 1, that sequence satisfies the hypothesis of the theorem, and the construction above gives TP_2 in the theory of $\mathcal{S}_\varphi$. The theory is therefore not simple, proving the assertion by contraposition. □

The array that witnesses the tree property of the second kind in the proof of Theorem 6.2 lies in an $\aleph_1$-saturated elementary extension of $\mathcal{S}_\varphi$. In the next two sections we show that, for the Pauli observable, the array parameters witnessing the tree property of the second kind for the atomic formula, the response itself, can be actual preparations, and that, for the Rado observable, every array that witnesses it for a quantifier-free formula in a probe variable with preparation parameters contains preparation parameters from an elementary extension outside the original structure. For the Pauli array, every finite set of conditions along a branch is realized by an error of finite support, while the whole branch is realized by a pure probe already in the original probe space, whose error label may have infinite support (see Theorem 7.2(a)).

7. The Pauli observable: syndrome sectors and the first check to fire

For the Pauli observable, the linear relations among check labels allow an infinite array of actual mixtures to witness the tree property of the second kind for the response predicate itself. The construction uses the indicator of a syndrome sector, the set of idealized errors with specified commutation values against finitely many checks. Within each block of linearly independent checks, Alice draws a label uniformly from the coset of a chosen check modulo the span of its predecessors. The response is 1 when that check is the first to anticommute with Bob's error, and $1/2$ when an earlier check already anticommutes with it. The resulting row conditions are incompatible at a fixed gap even for mixed probes, whereas choices from finitely many rows are realized by a finite-support error when all the checks together are

linearly independent. Thus the array parameters are actual preparations; an entire infinite path is realized by a pure probe already in the original probe space, whose error label may have infinite support, as specified in Theorem 7.2(a).

7.1. **Syndrome sectors.** Let d be a positive integer, let $g_1, \dots, g_d$ be linearly independent checks in the space V of labels of Pauli strings of finite support, with span G, and let a syndrome $\sigma \in \mathbb{F}_2^d$ be given. We regard σ as the linear functional on G with $\sigma(g_i) = \sigma_i$, and we call the set $\Sigma_\sigma = \{\lambda \in V^* : \lambda(g_i) = \sigma_i \text{ for } i \le d\}$ of idealized errors with syndrome σ its *syndrome sector*. For $s \in G$ we write r_s for the pure response r_{δ_s}, the response of the point mass at s, and we recall that χ_s is the character of V^* given by $\chi_s(\lambda) = (-1)^{\lambda(s)}$.

Proposition 7.1. *Let d be a positive integer, let $g_1, \dots, g_d \in V$ be linearly independent checks with span G, and let $\sigma \in \mathbb{F}_2^d$ be a syndrome, regarded as the linear functional on G with $\sigma(g_i) = \sigma_i$ for $i \le d$. Then the indicator of the syndrome sector Σ_σ satisfies, at every pure probe $\lambda \in V^*$,*

$$\mathbf{1}_{\Sigma_\sigma}(\lambda) = 2^{-d} \sum_{s \in G} (-1)^{\sigma(s)} \chi_s(\lambda) = 2^{-d} \sum_{s \in G} (-1)^{\sigma(s)} - 2^{1-d} \sum_{s \in G} (-1)^{\sigma(s)} r_s(\lambda).$$

Consequently, if $\sigma \neq 0$, then $\mathbf{1}_{\Sigma_\sigma} = r_{\mu_\sigma^1} - r_{\mu_\sigma^0}$ on pure probes, where μ_σ^1 and μ_σ^0 are the uniform probability measures on $\{s \in G : \sigma(s) = 1\}$ and on $\{s \in G : \sigma(s) = 0\}$; and $\mathbf{1}_{\Sigma_0} = 1 - 2r_{\mu_G}$, where μ_G is the uniform probability measure on G. In particular, for every probe ν, the probability that an error drawn from the Pauli channel ν has syndrome σ is $r_{\mu_\sigma^1}(\nu) - r_{\mu_\sigma^0}(\nu)$ if $\sigma \neq 0$, and $1 - 2r_{\mu_G}(\nu)$ if $\sigma = 0$.

Proof. The map $s \mapsto (-1)^{\sigma(s)} \chi_s(\lambda) = (-1)^{(\sigma+\lambda)(s)}$ is a character of the finite group G; its sum over G is 2^d if the linear functional $\sigma + \lambda$ vanishes on G, that is, if $\lambda \in \Sigma_\sigma$, and 0 otherwise, since a nonzero functional on G takes the value 1 on exactly half of G. This is the first equality, and the second follows from $\chi_s = 1 - 2r_s$. If $\sigma \neq 0$, the first sum on the right vanishes, and the two level sets of σ have 2^{d-1} elements each; grouping the second sum by the value of σ gives $r_{\mu_\sigma^1} - r_{\mu_\sigma^0}$. If $\sigma = 0$, the first sum is 2^d, and the second is $2^d r_{\mu_G}$. Both sides of each identity are affine in the probe, and integration against ν, together with Proposition 5.4(b), yields the last statement. □

When the checks commute pairwise and carry signs, the operator $2^{-d} \sum_{s \in G} (-1)^{\sigma(s)} g_s$, where g_s is the element with label s of the group generated by the signed checks, is the projector onto the states of syndrome σ [Got97], and the probability of the outcome σ when the checks are measured on a code state corrupted by a Pauli channel ν is given by Proposition 7.1. The proposition requires no commutation: the checks need only be linearly independent. The coefficients of the indicator, as an affine combination of Alice's responses, have total absolute value at most $1 + 2 = 3$, whatever the number d of checks.

7.2. **The first check to fire.** The next theorem arranges the syndrome sectors of the initial segments of blocks of checks in an array that witnesses the tree property of the second kind.

Theorem 7.2 (The first check to fire). *Let $(g_{im})_{i,m \in \mathbb{N}}$ be linearly independent vectors of V, put $W_{ij} = \operatorname{span}(g_{im} : m < j)$ for $i, j \in \mathbb{N}$, and let μ_{ij} be the uniform probability measure on the coset $g_{ij} + W_{ij}$. For the Pauli observable, the following hold:*

(a) *for every $f : \mathbb{N} \to \mathbb{N}$ and every $n \in \mathbb{N}$, there is an error $e \in V$ with $r_{\mu_{if(i)}}(e) = 1$ for all $i < n$, and there is an idealized error $\lambda \in V^*$ with $r_{\mu_{if(i)}}(\lambda) = 1$ for all i;*

(b) *for every* i, *every* $j \neq j'$ *and every probe* $\nu \in \mathcal{P}(V^*)$, *we have* $\min\big(r_{\mu_{ij}}(\nu), r_{\mu_{ij'}}(\nu)\big) \leq 2/3$, *and the bound* $2/3$ *is attained.*

Consequently, the family of responses of Alice has the tree property of the second kind in the sense of Definition 5.2, with $k = 2$, $\alpha = 2/3$ *and* $\beta = 1$.

Proof. We first compute the response of μ_{ij} at a pure probe $\lambda \in V^*$. If λ does not vanish on W_{ij}, then the functional λ takes the value 1 on exactly half of the coset $g_{ij} + W_{ij}$, since a nonzero functional on a finite $\mathbb{F}_2$-space is balanced, and translation by g_{ij} either preserves or exchanges its level sets. If λ vanishes on W_{ij}, then the functional λ is constant on the coset, with value $\lambda(g_{ij})$. Hence

$$r_{\mu_{ij}}(\lambda) = \begin{cases} 1 & \text{if } \lambda(g_{im}) = 0 \text{ for } m < j \text{ and } \lambda(g_{ij}) = 1, \\ 1/2 & \text{if } \lambda(g_{im}) = 1 \text{ for some } m < j, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

In words, the response is 1 exactly when g_{ij} is the first check of the block $(g_{im})_{m\in\mathbb{N}}$, which we call block i, to fire, that is, to take the value 1 under λ.

(a) Let f and n be given. The conditions $\lambda(g_{im}) = 0$ for $m < f(i)$ and $\lambda(g_{if(i)}) = 1$, for $i < n$, concern finitely many linearly independent vectors, so Lemma 3.2 gives an error e with $\omega(\cdot, e)$ satisfying them, and by (1) the response $r_{\mu_{if(i)}}(e)$ equals 1 for all $i < n$. The conditions for all i together define a nonempty closed subset of V^*, since the vectors g_{im} are linearly independent; any λ in it works.

(b) By symmetry, we may assume that $j < j'$. By (1), the pair $\big(r_{\mu_{ij}}(\lambda), r_{\mu_{ij'}}(\lambda)\big)$ at a pure probe takes one of the values

$$(1, \tfrac{1}{2}), \quad (\tfrac{1}{2}, \tfrac{1}{2}), \quad (0, \tfrac{1}{2}), \quad (0, 1), \quad (0, 0).$$

Indeed, if the first check of block i to fire among $g_{i0}, \dots, g_{ij'}$ has index m, the pair is $(\frac{1}{2}, \frac{1}{2})$ for $m < j$, $(1, \frac{1}{2})$ for $m = j$, $(0, \frac{1}{2})$ for $j < m < j'$ and $(0, 1)$ for $m = j'$; if none fires, the pair is $(0, 0)$. Responses are affine in the probe, so at a mixed probe ν the pair lies in the convex hull of these five points. Every listed pair (s, t) satisfies $s + 2t \leq 2$, and the same inequality holds throughout their convex hull. Hence $3\min(s, t) \leq s + 2t \leq 2$, so the minimum of the two coordinates is at most $2/3$. Equality requires $s = t = 2/3$, and this point is $\frac{2}{3}(1, \frac{1}{2}) + \frac{1}{3}(0, 1) = (\frac{2}{3}, \frac{2}{3})$. This point is the pair of responses at the probe $\frac{2}{3}\delta_{\lambda_1} + \frac{1}{3}\delta_{\lambda_2}$, where λ_1 and λ_2 are idealized errors under which g_{ij} and $g_{ij'}$, respectively, are the first checks of block i to fire; such idealized errors exist by (a).

Statements (a) and (b) are the two clauses of Definition 5.2 for the array $(r_{\mu_{ij}})$, with $k = 2$, $\alpha = 2/3$ and $\beta = 1$. □

Remark 7.3. For qudits of prime dimension p, the quantum systems with Hilbert space $\mathbb{C}^p$, the labels form a symplectic space over the field $\mathbb{F}_p$ with p elements, and the response of a label at an error is 1 when their commutation phase, the pth root of unity by which the products of the two strings in the two orders differ, is nontrivial and 0 otherwise. The proof of Theorem 7.2 adapts as follows. A nonzero functional on a finite $\mathbb{F}_p$-space takes each value equally often, so in (1) the conditions $\lambda(g_{ij}) = 1$ and $\lambda(g_{im}) = 1$ become $\lambda(g_{ij}) \neq 0$ and $\lambda(g_{im}) \neq 0$, and the response $\frac{1}{2}$ becomes $\vartheta = \frac{p-1}{p}$. For $j < j'$, the pair of responses at a pure probe takes one of the values $(1, \vartheta)$, (ϑ, ϑ), $(0, \vartheta)$, $(0, 1)$ and $(0, 0)$. The point (ϑ, ϑ) lies below the segment from $(1, \vartheta)$ to $(0, 1)$, since $\vartheta < 1 - \vartheta + \vartheta^2$, and on that segment the

two coordinates are equal at the point with first coordinate $\frac{p}{p+1}$. Hence the bound $\frac{2}{3}$ of Theorem 7.2(b) becomes $\frac{p}{p+1}$, and it is attained.

In the notation of Theorem 7.2 and the language of Proposition 7.1, the event "the check g_{ij} fires first in block i" is the syndrome sector of the checks $g_{i0}, \dots, g_{ij}$ with syndrome $(0, \dots, 0, 1)$, whose two level sets are the coset $g_{ij} + W_{ij}$ and the group W_{ij}. Hence its indicator is $r_{\mu_{ij}} - r_{\mu_{ij}^0}$ on pure probes, where μ_{ij}^0 is the uniform probability measure on W_{ij}. Only the linear independence of the checks enters Theorem 7.2; when the checks commute pairwise, each W_{ij} is the set of labels of a stabilizer group, and μ_{ij} draws uniformly from a coset of it. The simplest instance is the repetition code.

Example 7.4. Index the qubits by pairs $(i, m) \in \mathbb{N}^2$, and for each $i \in \mathbb{N}$ call the qubits (i, m), $m \in \mathbb{N}$, the chain i; let $g_{im} = Z_{i,m}Z_{i,m+1}$ be the check of the bit-flip repetition code that compares the neighboring qubits m and $m + 1$ of the chain i, where $X_{i,m}$ and $Z_{i,m}$ denote the bit flip and the phase flip on the qubit (i, m). These checks commute pairwise and are linearly independent. For a bit flip $e = \prod_{i,m} X_{i,m}^{x_{i,m}}$, with exponents $x_{i,m} \in \mathbb{F}_2$, we have $\omega(g_{im}, e) = x_{i,m} + x_{i,m+1}$, which is 1 if and only if exactly one of the qubits m and $m + 1$ of the chain i is flipped, that is, if and only if e has a domain wall between them. By (1), the response at e of the uniform probability measure μ_{ij} on the coset of g_{ij} modulo the span of $g_{i0}, \dots, g_{i,j-1}$, as in Theorem 7.2, is 1 exactly when the first domain wall of e on the chain i lies between the qubits j and $j + 1$. For this code, Theorem 7.2 reads as follows: the chains are independent code blocks of a memory, an error cannot have two first domain walls on the same chain, and the positions of the first domain walls on finitely many chains can be prescribed independently by a single error. The bit flips on the chain i with zero syndrome are the identity and the flip of every qubit of the chain, which acts on infinitely many qubits: the logical bit flip of an infinite chain is an idealized error, an element of $\hat{V} \setminus V$.

Remark 7.5. The mechanism behind Theorem 7.2 is Fourier analysis on the compact group V^*: since the characters of V^* separate points and are closed under multiplication, the linear span of Alice's responses and the constants is dense in $C(V^*)$ by the Stone–Weierstrass theorem; it is an algebra, because $2r_a r_b = r_a + r_b - r_{a+b}$ for the pure responses $r_a = r_{\delta_a}$, $a \in V$. By Proposition 7.1, the indicators of syndrome sectors, which are functions of supremum norm one, lie in this span with coefficients of total absolute value at most 3. In the first-order structure (V, ω), by contrast, a formula speaks about a bounded number of labels, and, by Theorem 3.3, no formula has the tree property.

The first-order theory of the finite-support symplectic label structure (V, ω) is simple and unstable (see Theorem 3.3); the real-valued theory of the preparation-and-probe structure $\mathcal{S}_\omega$ has the tree property of the second kind. In the latter structure a preparation can average over a finite set of labels whose size grows along the array. The coset mixtures of Theorem 7.2 make this distinction explicit: the atomic response predicate already witnesses the tree property with actual preparation parameters, strengthening the difference-of-responses construction of Theorem 6.2. The description of nonforking by intersections of spans and the gluing of compatible syndromes concern the label structure (see Theorem 3.3 and Proposition 4.8); the mixed response theory is a different object.

8. The Rado observable: marginals and idealized preparations

For the Rado observable, the mixed theory has the same tree property as for Pauli, but the preparation parameters required to witness it are different. The pure probe space is the full cube of adjacency patterns, and the response of an actual mixture depends only on the probabilities with which a probe declares individual vertices to be neighbors. The freedom to

prescribe these marginals separately on disjoint blocks prevents an infinite uniformly inconsistent row with actual mixture parameters, first for bounded affine combinations and then for each fixed continuous connective applied to finitely many responses (see Theorems 8.1 and 8.3). Consequently, a witnessing array for a quantifier-free formula of this form, in a probe variable with preparation parameters, must contain preparation parameters from an elementary extension outside the original structure. The restriction concerns the parameters of the array; the probes in the compatibility results range over the full probe space, including idealized probes. The same argument applies to the random bipartite graph, whose two vertex classes serve as Alice's and Bob's labels.

8.1. **Affine combinations of responses.** Consider a diagonal observable with freely independent pure responses, that is, one for which every pattern $\lambda \in \{0,1\}^A$ of answers to Alice's labels, the elements of A, is an idealized label of Bob. The response of a mixture depends on a probe ν only through its marginals $\pi_a(\nu) = \nu\{\lambda : \lambda(a) = 1\}$, for $a \in A$, as for the Rado observable in Section 5.5, and every vector of marginals in $[0,1]^A$ is attained by a product measure. A bounded affine combination of responses is therefore a function

$$h_{c,\gamma}(\nu) = \gamma + \sum_{a \in A} c(a)\, \pi_a(\nu), \qquad \gamma \in \mathbb{R},\ c \in \ell^1(A),$$

and its supremum over the probe space is $\gamma + \sum_a c(a)^+$, attained at the pure probe $\lambda = \mathbf{1}_{\{c>0\}}$. Two such functions can conflict only through the labels on which both put weight. In the proof of Theorem 8.1 below we show, by a gliding hump, that a sequence of such functions along which $|\gamma| + \|c\|_1$ stays bounded has a subsequence whose members put their weight, up to small errors, on a common finite set of labels and on pairwise disjoint finite blocks; for every positive integer k, a single pure probe then nearly maximizes k of them.

Theorem 8.1 (Compatibility of affine conditions). *Let C_φ be the diagonal observable of a function $\varphi\colon A \times B \to \{0,1\}$ on countable sets A and B, with freely independent pure responses, and let $\kappa > 0$ be given. Let $\mathcal{R}_\kappa$ be the family of functions $\gamma + \sum_{i<n} c_i r_{\mu_i}$ on the probe space $\mathcal{P}(\{0,1\}^A)$ of Radon probability measures on $\{0,1\}^A$, where $n \in \mathbb{N}$, the μ_i are mixtures of Alice, and γ and the c_i are reals with $|\gamma| + \sum_i |c_i| \le \kappa$. Then the family $\mathcal{R}_\kappa$ contains no uniformly k-inconsistent sequence, for any integer $k \ge 2$. In particular, no tree property is witnessed in $\mathcal{R}_\kappa$.*

Proof. Let $(h_j)_{j\in\mathbb{N}}$ be a sequence in $\mathcal{R}_\kappa$ and let $\alpha < \beta$ be such that each h_j takes a value $\ge \beta$. We show that for every integer $k \ge 2$ there are $j_1 < \cdots < j_k$ and a pure probe λ with $h_{j_\ell}(\delta_\lambda) > \alpha$ for $\ell = 1, \dots, k$.

Write $h_j = \gamma_j + \sum_i c_{j,i} r_{\mu_{j,i}}$, with real coefficients $c_{j,i}$ and mixtures $\mu_{j,i}$ of Alice; then $h_j = h_{c_j,\gamma_j}$ with $c_j = \sum_i c_{j,i}\mu_{j,i}$, whence $\|c_j\|_1 \le \kappa$. Since A is countable, a diagonal argument gives a subsequence, still denoted (c_j), that converges pointwise on A to some c; by Fatou's lemma, the limit satisfies $\|c\|_1 \le \kappa$. Put $\eta = (\beta - \alpha)/5$ and choose a finite set $F \subseteq A$ with $\|c \restriction (A \setminus F)\|_1 < \eta$. Put $F_0 = F$, and choose inductively indices $j_1 < \cdots < j_k$ and finite sets $F_0 \subseteq F_1 \subseteq \cdots \subseteq F_k$ such that

$$\|(c_{j_\ell} - c) \restriction F_{\ell-1}\|_1 < \eta \quad \text{and} \quad \|c_{j_\ell} \restriction (A \setminus F_\ell)\|_1 < \eta;$$

the first condition holds for all large indices by pointwise convergence, and the second for a large enough finite set. Define $\lambda \in \{0,1\}^A$ by $\lambda(a) = 1$ exactly when $a \in F$ and $c(a) > 0$, or $a \in F_\ell \setminus F_{\ell-1}$ and $c_{j_\ell}(a) > 0$ for some $\ell \le k$.

Let $\ell \leq k$ be given. The deficit of λ for h_{j_ℓ} is

$$\sup h_{j_\ell} - h_{j_\ell}(\delta_\lambda) = \sum_{\lambda(a)=0} c_{j_\ell}(a)^+ + \sum_{\lambda(a)=1} c_{j_\ell}(a)^-,$$

and we bound its contribution on four disjoint pieces of A. On $F_\ell \setminus F_{\ell-1}$ the sign pattern of λ is that of c_{j_ℓ}, and the contribution is 0. On F it is that of c, and a label where the signs of c_{j_ℓ} and c disagree contributes at most $|c_{j_\ell}(a) - c(a)|$; the total is less than η. On $F_{\ell-1} \setminus F$ the contribution is at most $\|c_{j_\ell} \restriction (F_{\ell-1} \setminus F)\|_1 \leq \|(c_{j_\ell} - c) \restriction F_{\ell-1}\|_1 + \|c \restriction (A \setminus F)\|_1 < 2\eta$. On $A \setminus F_\ell$ it is less than η. Hence $h_{j_\ell}(\delta_\lambda) > \sup h_{j_\ell} - 4\eta \geq \beta - 4\eta > \alpha$. □

Corollary 8.2. *Let $n \in \mathbb{N}$ be given, and let γ and c_i, for $i < n$, be reals. For the Rado observable, no tree property of the affine formula $\gamma + \sum_{i<n} c_i R(y_i, x)$, in which R is the response predicate, the variables y_i are of the sort of preparations and the variable x is of the sort of probes, is witnessed by actual preparations, although the real-valued theory of the structure $\mathfrak{S}_E$ of its mixed preparations and probes has the tree property of the second kind.*

Proof. By Lemma 5.5, the Rado observable has freely independent pure responses. With actual preparations as parameters, the instances of the formula lie in the family $\mathcal{R}_\kappa$ of Theorem 8.1 for every $\kappa > 0$ with $\kappa \geq |\gamma| + \sum_{i<n} |c_i|$, and that theorem yields the first statement. The second statement is Theorem 6.2. □

The distinction between finite and infinite arrays is essential here. For the fixed difference-of-responses formula of Theorem 6.2, actual mixtures provide arrays of every finite size, but Theorem 8.1 excludes an infinite uniformly inconsistent row whose preparation parameters are all actual mixtures. Hence each infinite row supplied by compactness must contain an idealized preparation parameter; the arbitrarily large finite arrays do not assemble into such a row inside the original structure. This conclusion also holds for the random bipartite graph, since its finite extension property makes every adjacency pattern a pure probe.

8.2. **Quantifier-free formulas.** In Theorem 8.1 the conditions on a probe are given by affine combinations of responses. In the real-valued logic of the second paper, a quantifier-free formula in a variable x of the sort of probes, built from the response with parameters of the sort of preparations, is a continuous function of finitely many responses, since the responses take values in $[0, 1]$ and the connectives are continuous functions. The next theorem is the analogue of Theorem 8.1 for these formulas. Its proof uses the same gliding hump, with one change: a continuous function of responses is no longer maximized by reading off signs (we show this after the proof), so instead of building the probe from signs we glue maximizers of the individual instances on disjoint blocks of labels.

Theorem 8.3 (Compatibility of quantifier-free conditions). *Let C_φ be the diagonal observable of a function $\varphi\colon A \times B \to \{0, 1\}$ on countable sets A and B, with freely independent pure responses, let $n \geq 1$ be an integer, and let $u\colon [0, 1]^n \to \mathbb{R}$ be a continuous function. Let $\mathcal{R}_u$ be the family of functions*

$$h(\nu) = u\big(r_{\mu_1}(\nu), \ldots, r_{\mu_n}(\nu)\big) \qquad (\nu \in \mathcal{P}(\{0, 1\}^A)),$$

where $\mu_1, \ldots, \mu_n$ range over the mixtures of Alice. Then the family $\mathcal{R}_u$ contains no uniformly k-inconsistent sequence, for any integer $k \geq 2$. In particular, no tree property

of the quantifier-free formula $u(R(y_1,x),\dots,R(y_n,x))$, where R is the response predicate, the variable x is of the sort of probes and the variables $y_1,\dots,y_n$ are of the sort of preparations, is witnessed by actual preparations.

Proof. For $\nu \in \mathcal{P}(\{0,1\}^A)$ write $\pi(\nu) \in [0,1]^A$ for its vector of marginals $(\pi_a(\nu))_{a\in A}$, defined before Theorem 8.1. Every $h \in \mathcal{R}_u$ factors as $h(\nu) = H(\pi(\nu))$ with

$$H(q) = u\big(\langle \mu_1, q\rangle, \dots, \langle \mu_n, q\rangle\big), \qquad \langle \mu, q\rangle = \sum_{a\in A} \mu(a) q_a,$$

and every $q \in [0,1]^A$ is the vector of marginals of a product measure. It therefore suffices to work on the cube $[0,1]^A$.

Let $(h_j)_{j\in\mathbb{N}}$ be a sequence in $\mathcal{R}_u$, with parameters $\mu_{j,1},\dots,\mu_{j,n}$ and associated functions H_j on the cube, and let $\alpha < \beta$ be such that each h_j takes a value $\geq \beta$. We show that for every integer $k \geq 2$ there are $j_1 < \cdots < j_k$ and $q \in [0,1]^A$ with $H_{j_\ell}(q) > \alpha$ for $\ell = 1,\dots,k$. Since u is uniformly continuous, there is $\theta > 0$ such that $|u(s) - u(t)| < \beta - \alpha$ whenever $\max_i |s_i - t_i| < \theta$. For each j choose $q^{(j)} \in [0,1]^A$ with $H_j(q^{(j)}) \geq \beta$. The set A is countable and the cube is compact and metrizable, so a diagonal argument provides a subsequence, still indexed by j, along which $\mu_{j,i} \to \mu_i$ pointwise on A for each $i \leq n$ and $q^{(j)} \to q^\infty$ in $[0,1]^A$. By Fatou's lemma, each limit μ_i has total mass at most 1. Put $\eta = \theta/5$ and choose a finite set $F \subseteq A$ with $\mu_i(A \setminus F) < \eta$ for all $i \leq n$.

Put $F_0 = F$, and choose inductively indices $j_1 < \cdots < j_k$ and finite sets $F_0 \subseteq F_1 \subseteq \cdots \subseteq F_k$ such that, for all $i \leq n$,

$$\sum_{a\in F_{\ell-1}} |\mu_{j_\ell,i}(a) - \mu_i(a)| < \eta, \qquad \sum_{a\in F} |q_a^{(j_\ell)} - q_a^\infty| < \eta, \qquad \mu_{j_\ell,i}(A \setminus F_\ell) < \eta.$$

The first two conditions hold for all large indices, by convergence on the finite sets $F_{\ell-1}$ and F, and the third for a large enough finite set F_ℓ. Define $q \in [0,1]^A$ by $q_a = q_a^\infty$ for $a \in F$, $q_a = q_a^{(j_\ell)}$ for $a \in F_\ell \setminus F_{\ell-1}$ and $\ell \leq k$, and $q_a = 0$ for $a \notin F_k$.

Let $\ell \leq k$ and $i \leq n$ be given. We compare $\langle \mu_{j_\ell,i}, q\rangle$ with $\langle \mu_{j_\ell,i}, q^{(j_\ell)}\rangle$, on the same four pieces of A as in the proof of Theorem 8.1; on each piece the two vectors differ by at most 1 in each coordinate. On $F_\ell \setminus F_{\ell-1}$ they agree. On F the difference is at most $\sum_{a\in F} |q_a^\infty - q_a^{(j_\ell)}| < \eta$, since $\mu_{j_\ell,i}(a) \leq 1$. On $F_{\ell-1} \setminus F$ it is at most $\mu_{j_\ell,i}(F_{\ell-1} \setminus F) \leq \sum_{a\in F_{\ell-1}} |\mu_{j_\ell,i}(a) - \mu_i(a)| + \mu_i(A \setminus F) < 2\eta$. On $A \setminus F_\ell$ it is at most $\mu_{j_\ell,i}(A \setminus F_\ell) < \eta$. Hence $|\langle \mu_{j_\ell,i}, q\rangle - \langle \mu_{j_\ell,i}, q^{(j_\ell)}\rangle| < 4\eta < \theta$ for every $i \leq n$, and the choice of θ gives $H_{j_\ell}(q) > H_{j_\ell}(q^{(j_\ell)}) - (\beta - \alpha) \geq \alpha$. □

For affine combinations, the proof of Theorem 8.1 yields slightly more: there the common witness can be taken pure. For an arbitrary connective u the witness is the product probe whose vector of marginals is the point q constructed in the proof of Theorem 8.3, and it cannot always be taken pure: for $n = 1$, $\mu = \delta_a$ with $a \in A$, and $u(t) = 1 - |2t - 1|$, the instance vanishes at every pure probe and equals 1 at the probes ν with $\pi_a(\nu) = 1/2$.

The contrast between the actual-parameter arrays is governed by the relations among pure responses. For Rado, the pure probe space is the full cube $\{0,1\}^\Gamma$, so marginals on disjoint blocks of labels can be prescribed independently. For each fixed quantifier-free formula in a probe variable with actual preparation parameters, the gliding-hump argument uses this freedom to extract arbitrarily large finite collections of compatible conditions, allowing any prescribed positive decrease of the common threshold. For Pauli, the pure probes are the linear functionals in V^*, and the relations among label coordinates place the indicators of

syndrome sectors in the affine span of responses with a coefficient bound independent of the number of checks (see Proposition 7.1). These identities support the actual-parameter array of Theorem 7.2, while Theorem 6.2 establishes the tree property of the mixed theory in both cases.

9. Affine logic and correlated preparations

Finally, we place the results in affine logic, in the form developed by Ben Yaacov, Ibarlucía and Tsankov [BYIT24], whose connectives are the affine maps and in which a mixture of structures is a direct integral. Formula values in a direct integral are integrals of formula values, which is the rule by which a response is computed on a mixed preparation. In this logic the type spaces are compact convex sets, and a structure is *extremal* when it realizes only extreme types, the extreme points of these sets. Ben Yaacov, Ibarlucía and Tsankov [BYIT24] showed that every model of an affine theory is an affine elementary substructure of a direct integral of extremal models. An affine theory is simplicial when its type spaces are Choquet simplices, that is, sets in which every point is the barycenter of exactly one probability measure that is maximal in the Choquet order; when the set is metrizable, the maximal measures are those concentrated on its extreme points. The same authors also showed that every model of a simplicial theory is such a direct integral, uniquely.

Ben Yaacov and Ibarlucía [BYI26] proved that an affine formula stable in almost every member of a measurable field of structures is stable in its direct integral, and that an affine theory is stable exactly when its extremal models are stable. Their definition of stability of a formula in a structure is the double limit condition of the first paper of this series [DI26a]: for every two sequences of elements, one in each variable of the formula, the two iterated limits of its values agree whenever both exist. Mixing in the sense of this paper behaves differently with respect to simplicity: the first-order theories of the Pauli group and of the Rado graph are simple, whereas, by Theorem 6.2, the real-valued theories of their observables have the tree property of the second kind.

The distinction between the two forms of averaging is already visible in the expectation kernel. In a direct integral, the elements in both variables may depend on the same random parameter, and the formula value is averaged over that common parameter; for a label relation, this permits correlated choices of Alice's and Bob's labels. Keisler's randomizations [Kei99, BY13] likewise use random elements defined on a common probability space. In the observable structure considered here, Alice's label and Bob's pure probe are drawn independently from their respective laws, and the response is their product average (see Lemma 5.1). The tree property established in Theorem 6.2 therefore already occurs with independent product preparations.

10. Questions

Question 10.1. Which diagonal observables with the independence property have the tree property of the second kind witnessed by actual preparations? For the diagonal observable C_φ of a function $\varphi\colon A \times B \to \{0,1\}$ on countable sets A and B, Proposition 7.1, Theorem 7.2 and Theorem 8.1 suggest a criterion in terms of the linear span of its responses in the space $C(K_\varphi)$ of continuous functions on the space K_φ of idealized labels of Bob: whether it contains, with coefficients of bounded total absolute value, the indicators of the sets of an infinite array of clopen sets whose rows consist of pairwise disjoint sets and in which every choice of one set from each of finitely many rows has nonempty intersection.

Question 10.2. By Remark 7.3, the proof of Theorem 7.2 adapts to qudits of prime dimension. For continuous variables, that is, for systems such as a particle on a line, whose position and momentum take real values, the commutation phase of a translation by a distance x and a momentum kick by $\hbar k$, where $\hbar$ is the reduced Planck constant and the wavenumber k is a real number measured in the inverse of the unit of length, is $e^{\mathrm{i}kx}$, whose real part $\cos kx$ describes an interference fringe. By the second paper, the fringe observable, which asks whether a photon lands on a bright fringe of an interference pattern of wavenumber 8^n, for a label $n \in \mathbb{N}$ of Alice, is not NIP. Where do interference fringes of unbounded wavenumber lie with respect to simplicity?

Question 10.3. In first-order model theory, a simple theory without the independence property is stable. Is an observable stable when its real-valued theory is simple? By Theorem 6.2, every simple diagonal observable is NIP, so for diagonal observables the question is whether those of the second paper that are NIP but unstable, such as the comparison of photon numbers, which asks whether Bob's system has more photons than Alice's, have a tree property once preparations are mixed.

Question 10.4. Does Theorem 6.2 hold for every observable whose responses have the independence property, not only for diagonal ones? The proof uses the values 0 and 1 of pure responses; for general observables they are replaced by two distinct levels, and the averages must be controlled with a gap.

Question 10.5. Is there an analogue of the Kim–Pillay theorem for affine logic, characterizing simple affine theories by an independence relation on extremal types?

Historical remarks

Simple theories were introduced by Shelah [She80] in 1980, together with the tree properties of the first and second kind; the motivating example was the theory of the countable random graph (see [She80, GIL02]). Kim [Kim98] proved in 1998 that forking is symmetric in simple theories, and Kim and Pillay [KP97] proved the independence theorem and the characterization of simplicity by an independence relation in 1997. Ben Yaacov [BY03] extended the theory in 2003 to compact abstract theories, a framework that contains the continuous logic of metric structures. The Rado graph was described by Ackermann [Ack37] in 1937 and by Rado [Rad64] in 1964; by a theorem of Erdős and Rényi [ER63] from 1963, it is the almost sure isomorphism type of the random graph on countably many vertices. Fraïssé [Fra54] introduced his limits in 1954, and Witt's theorem on isometries of forms is in Artin's *Geometric Algebra* [Art57]. Vector spaces of countably infinite dimension over finite fields with a nondegenerate alternating form, such as the Pauli group, are among the Cherlin–Hrushovski smoothly approximable structures [CH03]. Granger [Gra99] studied vector spaces with a generic bilinear form over an arbitrary field in his thesis of 1999 and showed that, over an infinite field carried as a second sort, the theory is not simple; Chernikov and Ramsey [CR16] proved in 2016 that over an algebraically closed field it is NSOP_1: it lacks the 1-strong order property, as every simple theory does. The symplectic description of the Pauli group and of stabilizer codes is due to Gottesman [Got97] and to Calderbank, Rains, Shor and Sloane [CRSS98]; the expression of the projector onto the states of a given syndrome as an average over the stabilizer group, behind Proposition 7.1, is standard in Gottesman's stabilizer formalism. Keisler [Kei99] introduced randomizations in 1999; Ben Yaacov and Keisler [BYK09] treated them as metric structures, and Ben Yaacov [BY13] proved in 2013 that the Keisler randomization of a theory with the independence property has TP_2. Direct integrals and extremal models in affine logic are due to Ben Yaacov, Ibarlucía and Tsankov

[BYIT24], and stability in affine logic to Ben Yaacov and Ibarlucía [BYI26]. The first two papers of this series are [DI26a, DI26b].

Declarations

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